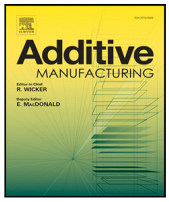
Additive Manufacturing 78 (2023) 103828



Contents lists available at ScienceDirect

Additive Manufacturing

journal homepage: www.elsevier.com/locate/addma



Research paper



An all-encompassing study on the joint effect of powder feedstock characteristics and manufacturing process parameters on the densification and mechanical properties of additively manufactured nickel alloy 718

Peter Morcos [a,1], David Shoukr [b,1], Tayler Sundermann [a], Thomas Dobrowolski [c], Nicholas Barta [c], Jayesh R. Jain [c], Raymundo Arróyave [a,b,d], Ibrahim Karaman [a], Alaa Elwany [a,b,*]

[a] Department of Materials Science and Engineering, Texas A&M University, College Station, TX, United States of America
[b] Wm Michael Barnes '64 Department of Industrial and Systems Engineering, Texas A&M University, College Station, TX, United States of America
[c] Baker Hughes, Houston, TX, United States of America
[d] J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, College Station, TX, United States of America

ARTICLE INFO

Keywords:
L-PBF
Nickel alloy 718
ANOVA
DOE

ABSTRACT

Laser-powder bed fusion (L-PBF) is a commonly employed additive manufacturing (AM) technique in manufacturing 3D metallic parts with complex geometries from powder materials. It is a complex process with more than 100 process parameters affecting the properties of fabricated parts. Studies on AM of nickel alloy 718 are currently focusing on investigating the effect of the process parameters and heat treatment on final part properties. Nevertheless, the effect of different powder characteristics such as particle size distribution (PSD), the presence of fine particles, powder reuse, and moisture content has not been studied comprehensively in the literature. In this study, two types of powder: virgin and recycled, as well as four parameters: PSD, which is represented by the median particle size and the particle size range, moisture content, and volumetric energy density (VED), were studied. To reduce experimental cost and burden, statistical design of experiment (DOE) techniques were utilized. Then, the measured results were analyzed using analysis of variance (ANOVA) to ascertain the effect of the different powder characteristics on multiple quantities of interest (QoIs). Different densification and mechanical tests were performed, and their outputs were studied as the QoIs. This study revealed that VED, powder type, PSD range, and median are the most common significant factors for most QoIs. Meanwhile, moisture content had a significant effect only on the Charpy impact energy. Besides, the study showed that the presence of fine particles < 20 μm improved the densification of the fabricated parts but reduced the mechanical properties.

1. Introduction

Laser-powder bed fusion (L-PBF) is a metal additive manufacturing (AM) process that has witnessed notable advances, particularly over the past decade [1]. In L-PBF, 3D parts are manufactured through the iterative deposition of powder layers using a blade or a roller then a laser beam selectively melts powder layers guided by a CAD file to form the final part. Different metallic materials and alloys have been successfully processed using L-PBF, including copper [2], 316L stainless steel (SS) [3], 15-5 PH SS [4], 17-4 PH SS [5], 321 SS [6], 420 SS [7], AlSi10Mg [8], Ti6Al4V [9], NiTi [10], AlMgScZr [11], WE43 magnesium alloy [12], W based alloys [12,13], nickel alloy 625 [14], and nickel alloy 718 [15]. The focus of the current work is on nickel alloy 718; a nickel-based superalloy characterized with excellent properties at elevated temperatures, making it an excellent candidate in high-temperature load-bearing and corrosive environments [16]. Nickel-based superalloys are among the most common materials processed using metal AM, with nickel alloy 718 being the most studied alloy, accounting for approximately 68% of literature studies according to Sanchez et al. [17].

The L-PBF process is complex, with as many as 100 process parameters affecting the properties of the final printed parts [18]. It has been generally established that the most influential parameters are laser power, scan speed, and hatch spacing. Additional process parameters that have been previously studied include substrate preheating temperature [19], layer thickness [7,20], oxygen content in the printing atmosphere, [21], and the printing atmosphere composition [22].

* Corresponding author.
E-mail address: elwany@tamu.edu (A. Elwany).
[1] The first two authors have contributed equally to this work.

https://doi.org/10.1016/j.addma.2023.103828
Received 5 March 2023; Received in revised form 9 October 2023; Accepted 13 October 2023
Available online 16 October 2023
2214-8604/© 2023 Elsevier B.V. All rights reserved.

In addition to the manufacturing processing parameters, it is well-established that the properties of the input raw feedstock (i.e., metallic powder) significantly affect part quality. A large body of the literature focuses on studying the effect of powder characteristics such as particle morphology, size distribution, and chemical composition. Farzadfar et al. [23] studied the mechanical properties of nickel alloy 718 samples fabricated from bi-modal and uni-modal coarse particle size distributions (PSD) and found that the latter are more ductile and have lower ultimate tensile strength and yield strength.

Meanwhile, Brika et al. [24] studied the influence of nickel alloy 718 particle morphology and PSD on powder flowability and mechanical properties of the fabricated samples. They noted that the presence of fine particles reduced powder flowability and that mechanical properties were directly proportional to the sphericity of the particles. The effect of powder reuse was also studied in some previous works. Gruber et al. [25] studied the change in the powder's PSD, particle shape, and flowability after reusing the powder for 20 cycles. The study revealed that the powder properties and hardness of the printed samples were not significantly affected. Cordova et al. [26] studied the effect of reusing the powder for 38 cycles on the powder size, morphology, oxygen content, and flowability. The average particle size increased, and particle morphology changed due to the presence of satellites. The flowability was slightly affected, and the oxygen content was not changed. Finally, Rock al. [27] studied the influence of powder reuse up to 10 cycles on powder size, morphology, oxygen content, and flowability of nickel alloy 718. They observed that as the number of cycles increased, the median particle size (D50), the flowability of the powder, and the oxygen content increased.

One important observation in the literature studies related to AM of nickel alloy 718 is that factors that influence the properties of printed parts have typically been studied in silos. For example, some studies emphasize the effect of manufacturing process parameters such as laser power, scan speed, layer thickness, or aggregate metrics of these parameters such as the linear energy density (LED) and volumetric energy density (VED). Other studies focus on analyzing the effect of powder feedstock characteristics. Furthermore, both directions consider very specific aspects, for example, the effect of powder recycling or the correlation between the aggregate VED and densification in the processed material. While all those studies are very important and have contributed to defining the state of the art in AM of nickel alloy 718, they still do not account for significant interactions and interplay between different factors.

The current work aims to fill the above gap by conducting a comprehensive study of the factors that influence the quality of AM-fabricated nickel alloy 718 and the interactions between these factors. Such analysis is lacking in the literature due to the specific focus of prior works or the costs associated with large experimental campaigns needed to draw meaningful and substantiated results. More specifically, we study the effect of (1) powder characteristics (particle size distribution, moisture content, and powder reuse) and (2) manufacturing processing parameters (represented by VED) on multiple quantities of interest (QoI). The QoIs are the powder-bed packing density, densification of printed cubes, tensile properties, and impact toughness. A systematic statistical design of experiments (DOE) and analysis of variance (ANOVA) are used to establish quantitative relationships between input factors and the QoIs. This involves extensive preparation of powder batches, manufacturing test coupons, mechanical test samples, and experimental characterization.

2. Experimental design

2.1. Down selection of parameters

In 2019, Cordova et al. [26] presented an in-depth analysis of the change in characteristics between virgin powder and recycled powder for multiple alloy systems. The study emphasized the use of a combination of standardized measurement techniques to quantify powder characteristics and design a reproducible characterization pathway to determine the efficacy of printing using recycled powders. A variant of the structure used in this study was used to describe pathways of analysis for metallic powder and is shown in Fig. 1.

Cordova et al.'s proposed characterization pathway is defined by chemical and physical morphology, though the modified characterization pathway used in this study features powder morphology, composition, and behavior. Powder morphology indicates a powder's physical characteristics, divided into size and shape. The analysis was achieved by measuring the PSD (volume, area, and normal distributions), median powder size, sphericity of powder, and presence of satellites. The composition was used rather than chemistry because it allowed a more accurate accounting of chemical and microstructural variations. As shown in Fig. 1, microstructure includes the phases and dendritic structure of powder particle microstructures and the porosity in the powder from gas entrapment. Chemical composition was measured for individual particles and the bulk material. These qualities combine to create a unique powder. However, since all of the previous measurements are static, and some are qualitative, dynamic testing must also be considered. Static measurements refer to the analysis that cannot describe how the powder moves, settles, or acts within a build environment. To compensate for a third branch, the behavior was analyzed as a combination of powder density, powder flowability, and the environmental reactivity of the powder. The authors will provide the details of this process in an upcoming publication.

After an extensive examination of the differences between virgin and recycled powders across multiple testing areas, encompassing various powder metrics such as particle size distribution, median particle size, sphericity, presence of satellite, microstructural phases, microporosity, particle chemistry, bulk chemistry, powder density, flowability, and environmental reactivity, the study revealed insignificant disparities between the two conditions. This observation strongly reinforces the concept that recycled powder printing is a feasible approach, as corroborated by numerous references in the literature covering diverse materials [28–30].

The chemical variance was within the bounds of measurement error and common specifications of nickel alloy 718. However, differences in the physical parameters, remelting that combines two particles into one, excess satellites, and non-uniform melting, were seen between the virgin and recycled powders. Nearly all observed inconsistencies were seen in non-standard particles. This indicates that if a way to segregate non-conforming recycled powder particles exists, the resulting recycled powders would be nearly indistinguishable from virgin ones. However, even though measurements indicated very few differences between virgin and recycled powders, it still needs to be investigated whether recycled powder can produce the same quality of build outcome as the virgin one.

The final selected parameters for investigation were powder type (recycled or virgin), PSD range (inclusion or exclusion of <20 μm particles), median particle size (D50), and moisture content (ambient or dried). Chemical composition was excluded as it appeared stable across all measured feedstocks. Powder size distribution was chosen to control the physical morphology of the powder. The analyzed feedstock powders differed in sphericity, size, and dynamic tests such as flowability. While testing can differentiate materials, manipulating them back to a common standard is not trivial or well understood, e.g., increasing the sphericity of powder A in comparison to powder B. However, PSD represents an easily measurable property and can be manipulated by sieving. Mixing sieved cuts of powder in such a way as to change the median particle size and include or exclude fine particles (<20 μm) yields four distinct distributions for each powder type. To capture variances that would not be removed by sieving and mixing, physical morphology was classified by referring to the original feedstock from which the manipulated distributions were mixed, i.e., recycled or

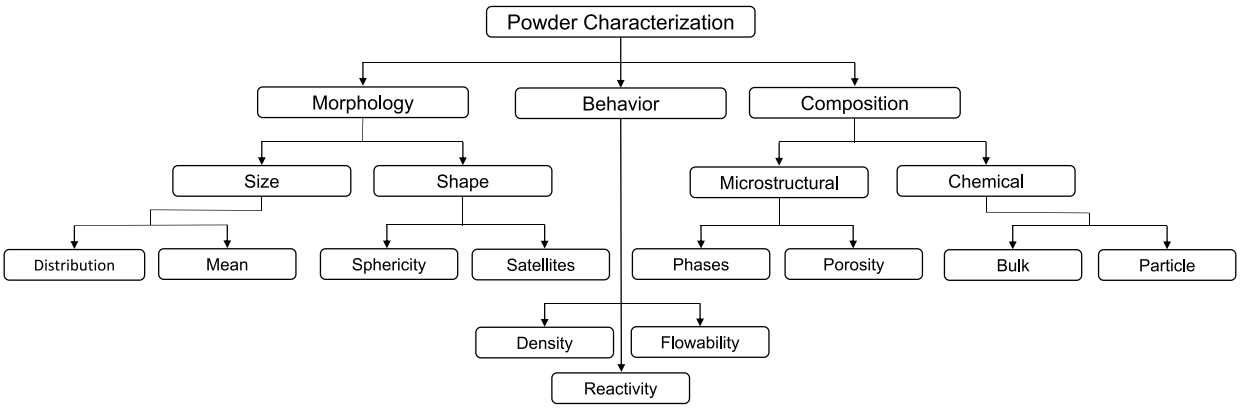


Fig. 1. Proposed characterization methodology for powder particles used in this study.

virgin. Moisture content was chosen to determine if special handling requirements of the feedstocks are required for industrial fabrication. Here, "ambient" refers to exposure to the air-conditioned environment of the laboratory, 22–25 °C and 50% reH (relative humidity). Meanwhile, the "dried" procedure requires heating powder in an inert kiln after the powder is loaded into machine interfacing containers but prior to machine insertion to reach 3%–5% reH.

Each down-selected parameter is denoted by a letter for easy referral in the rest of the study; A, B, C, D, and E. The powder type (Factor A), PSD (which includes the particle size range (Factor B), and the median particle size (Factor C)), moisture content (Factor D), and the VED (Factor E) are studied for their influence on the densification, properties, and performance of the final fabricated part. A DOE methodology is used to assess the significant effect of these factors and their interactions on the final AM part performance. In the following subsections, we discuss the selection of the experimental factors and their levels, the control of the factor levels, and the DOE in detail.

2.2. Selection of factor levels

Two levels were selected for each of the parameters under investigation. These levels were chosen to represent the expected variability range. They represent the possible categories or the lower and upper bounds of the selected factor range.

2.2.1. Powder type (factor A)

Powder type is represented by two types of powders, virgin powder (A1) and recycled powder (A2). The ability to use recycled powder is an important advantage of additive manufacturing [31]. This factor is intended to test the variation due to the use of recycled powder on the different QoI. The definition of recycled in this case is that virgin powder circulated at least once through the machine and was conditioned using particle size segregation (100 μm sieve mesh size) and dried below 5% reH before batch preparation.

The chemical composition of the virgin and recycled powder is shown in Table 1. All powder compositions meet the UNS N07718 in API standard 6ACRA [32]. The two types of recycled powder in Table 1 refer to the powder that is removed from the chamber on the first cleaning step (recycled far from the part) and the powder removed in subsequent steps (recycled close to the part). The recycled powder used in this study is a combination of these feedstocks; in previous work by the authors, it was determined that the differences between nickel alloy 718 virgin powder and recycled powder were primarily, if not exclusively, morphological and could be accounted for by sieving and mixing. Therefore, the physical characteristics were rigorously tested here, but features previously proven indifferent to recycling are not included.

Table 1
Bulk powder chemical composition for both virgin and recycled powder.

Element	Virgin [wt %]	Recycled (Far) [wt %]	Recycled (Close) [wt %]
Al	0.4382	0.4477	0.4552
B	<0.005	<0.005	<0.005
C	0.0341	0.0329	0.0335
Co	0.0623	0.0628	0.0612
Cr	19.6110	19.4090	19.7000
Cu	0.0138	0.0133	0.0133
Fe	17.9540	18.3970	18.5000
H	0.0004	0.0005	0.0005
Mn	0.0045	0.0041	0.0037
Mo	3.1181	3.0513	3.0982
Nb	5.0276	4.9763	5.0541
N	0.0034	0.0043	0.0054
Ni	52.6942	52.5714	52.0309
O	0.0130	0.0153	0.0152
P	<0.005	<0.005	<0.005
S	0.002	0.002	0.002
Si	0.0298	0.0282	0.0278
Ti	0.9736	0.9639	0.9790
V	0.02	0.02	0.02

2.2.2. Particle size distribution (factors B and C)

In this study, the PSD is modified by varying the particle size range and median (D50). This process is done by sieving each powder type through a tower of sieves of different sizes and then mixing each powder cut (the remaining powder between two sieve sizes) based on the calculations presented below and the calculations in Appendix A. Each combination of D50 and particle size range provides a unique PSD.

2.2.2.1. Particle size range (factor B). The typical particle size range of the nickel alloy 718 powder for additive manufacturing is from 15 to 63 μm [33]. Fine particles, with a size less than <20 μm, act as interstitial particles. They rest in the voids between coarse particles, which increases the theoretical powder-bed packing density during the additive manufacturing process [34] as shown in Fig. 2. On the other hand, fine particles have higher electrostatic forces, which tend to cause agglomeration, consequently reducing the flowability of the powder [35,36].

To test the effect of fine particles on the different QoIs, the two levels of the powder size range were selected such that one level does not include the fine particles (denoted by B1), i.e., has particle size with a lower bound ≥20 μm. In contrast, the other level has a full particle size range (denoted by B2). The upper bound of the size range is 75 μm based on the measured PSDs of both the virgin and recycled powders. The lower bound for level B1 (20 μm) is chosen based on the smallest standard sieve mesh size [38].

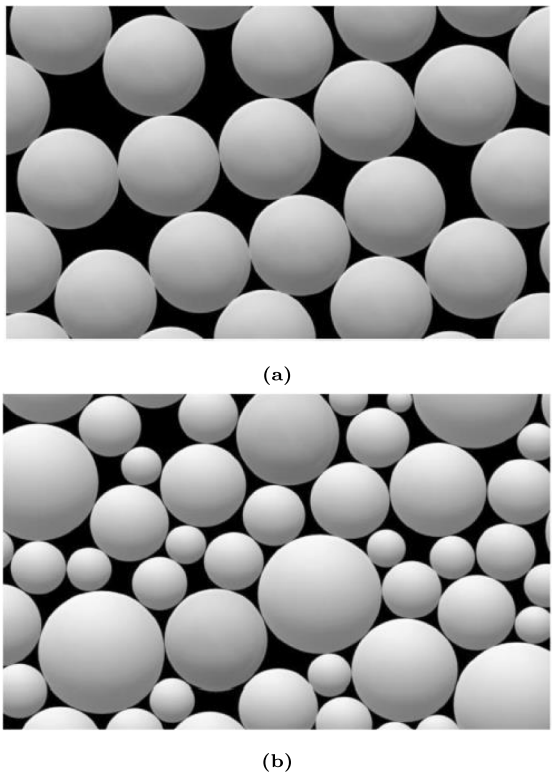


Fig. 2. A schematic representation for the powder bed packing (a) without interstitial particles, and (b) with interstitial particles [37].

2.2.2.2. Median particle size; D50 (factor C). The D50 values for nickel alloy 718 vary between 25 and 50 μm for additive manufacturing applications [35]. The D50 was controlled via a thorough sieving process followed by mixing of sieved powder cuts with certain ratios.

The two levels of D50 were determined via preliminary calculations based on the PSD of each powder type, namely 27 μm (C1) and 37 μm (C2). The two levels are the D50 of the original powder of 32.02 μm ±5 μm. A difference of 10 μm between the lower and upper values can clarify the variability because of the change in D50. Also, using a lower D50 level of less than 27 μm would significantly increase the sieving burden for the batch with a range of 20:75 μm with a D50 of 27 μm. Moreover, using a higher level of 37 μm reduces the amount of powder needed (a higher level will incur notable powder waste to control the D50 value). It is worth mentioning that despite an extensive sieving process, the sieved powder cuts will still include a small number of particles outside the nominal sizes of the lower and upper sieves.

Control of the median particle size (D50) levels and the optimization of the powder mixing process are described in Appendix A. The resultant PSDs of the virgin and recycled powder batches are shown in Figs. A.3 and A.4, respectively. Both figures show great conformity between the predicted and the mixing PSDs (by volume). The PSD by count is used to validate the amount of powder (by count) under 20 μm in size. For the batches with the range 0:75 μm, all the PSDs contained more than 45% (by count) powder particles of size <20 μm. Meanwhile, for the batches with the range 20:75 μm, all the PSDs contained at most 20% powder particles <20 μm, by count.

2.2.3. Moisture content (factor D)

Exposure to the ambient environment has been a constant issue in the development of additive manufacturing. For example, Ti6Al4V is a workhorse in additive manufacturing but suffers challenges associated with handling and recycling because it readily picks up oxygen, nitrogen, carbon, and hydrogen; inclusions which are detrimental to final part outcomes [39–41]. While nickel alloy 718 does not seem to have significant environmental reactivity, the effects of moisture content on the feedstock are of interest. Therefore, to determine if the exposure of the powder particles to the ambient atmosphere has an impact on the QoIs, two levels of moisture content are used; low moisture content level; or dried level (D1), and high moisture content level; or ambient level (D2).

A low moisture content level is considered to be nearly zero. After each batch was mixed to specification, it was sealed under argon at 102 kPa with residual oxygen and water contents below 0.5 PPM. Then, the powder was transferred to production tanks under an inert environment and baked in an autoclave to remove any trace moisture. On the other hand, high moisture content was simply obtained by the absence of rigorous steps; after mixing, the powder was sealed in the atmosphere with desiccant for shipping, upon receiving it was placed in a tray in the production environment that kept out impurities but would maximize air exposure for 24 h. It was then loaded into the machine. In theory, the two levels of the QoI compare a powder under ideal environmental conditions and one fully exposed.

2.2.4. Volumetric energy density (factor E)

The VED is calculated from the following equation:

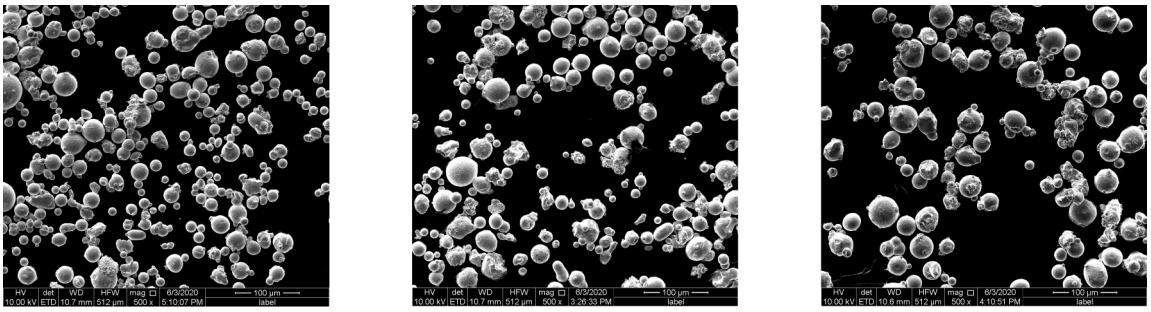
$$VED = \frac{P}{v * t * h} \tag{1}$$

where, P is the laser power, v is the scanning speed, h is the hatch spacing, and t is the layer thickness. Therefore it combines the effect of the four processing parameters of the L-PBF. The two VED levels selected in this study are 48.6 J/mm^3 (E1) and 36.1 J/mm^3 (E2). They represent the high and low levels, respectively. The value of VED 1 was chosen to be within the good printability region [20], while VED 2 was selected to be close to the region boundary. One of the objectives of this study is to evaluate changes in the sensitivity to powder characteristics at these locations. The latter has industrial benefits in certain situations despite inferior properties.

2.3. Design of experiment

Design of experiments is a statistical tool used to determine the effect of experimental factors and their interactions and ensure the experiment is well-balanced while minimizing the experimental cost [42]. A typical DOE utilizes full factorial designs. These designs include all possible combinations of factor levels, allowing for an exhaustive analysis of all factor interactions. While a full factorial design is informative, it is the most expensive of all the factorial designs. Moreover, according to the hierarchical ordering principle [42], higher-order interactions, i.e., three-factor interactions, are rarely significant. That means that the primary factors and the low-order interactions dominate the system [43]. This conforms with the sparsity-of-effect principle that only a few effects will be statistically significant. To that extent, fractional factorial designs are used. In these designs, only a fraction, $^1/_2$ or $^1/_3$, based on the number of factor levels, of the full factorial experiments are used. These experiments are selected based on a defining relation such that the highest (or one of the higher) order interaction has all plus levels, thus eliminated. An example of a defining relation is I = ABC, for which A = BC, B = AC, and C = AB, and ABC is eliminated.

In this study, a DOE based on a $^1/_2$ fractional factorial design is implemented. As discussed in Section 2.2, four of the five studied factors affect the powder preparation, while the VED factor levels can be performed on the same powder batch without added cost. Hence, a two-step experimental design is applied to maximize the information from the experiment while minimizing the cost. First, a $^1/_2$ fractional factorial DOE for the first four factors is established, which reduces the number of experiments from 2^4 to 2^{4-1} experiments. For which, the defining relation is **I = ABCD**. Second, these eight experiments, representing 8 different powder batches, are repeated for each of the two VED levels, which yields a total of 16 experiments that can be



(a) (b) (c)

Fig. 3. SEM images for the (a) virgin powder, (b) Recycled powder (Far), and (c) Recycled powder (Close).

Table 2
Experimental runs according to the selected DOE.

	Powder type (A)	Particle size range (B)	Median particle size (C)	Moisture content (D)	VED (E)
1	Virgin	20: 75 μm	27 μm	Dry	VED 2
2	Virgin	20: 75 μm	37 μm	Ambient	VED 2
3	Virgin	0: 75 μm	27 μm	Ambient	VED 1
4	Virgin	0: 75 μm	37 μm	Dry	VED 1
5	Recycled	20: 75 μm	27 μm	Ambient	VED 2
6	Recycled	20: 75 μm	37 μm	Dry	VED 2
7	Recycled	0: 75 μm	27 μm	Dry	VED 1
8	Recycled	0: 75 μm	37 μm	Ambient	VED 1
9	Virgin	20: 75 μm	27 μm	Dry	VED 1
10	Virgin	20: 75 μm	37 μm	Ambient	VED 1
11	Virgin	0: 75 μm	27 μm	Ambient	VED 2
12	Virgin	0: 75 μm	37 μm	Dry	VED 2
13	Recycled	20: 75 μm	27 μm	Ambient	VED 1
14	Recycled	20: 75 μm	37 μm	Dry	VED 1
15	Recycled	0: 75 μm	27 μm	Dry	VED 2
16	Recycled	0: 75 μm	37 μm	Ambient	VED 2

performed in 8 runs (VED can be changed during the same run). The actual experiments used in this study are shown in Table 2.

The resulting 1/2 fractional factorial DOE includes 16 experimental runs and can estimate the effect of 15 factors and interactions (as our DOE has 15 degrees of freedom), see Table 3. Moreover, this is a resolution IV design (**I** = **ABCD** ⇒ 4 letters). These designs have the following characteristics [44]:

1. The main effects are clear, i.e., no main effect is aliased with any two-factor interaction, and
2. Two-factor interactions are aliased with each other.

These characteristics can be seen in Table 3. Moreover, this experiment is designed such that factor E (VED) is not reduced. The reason for that, as mentioned before, is that VED can be varied across the same build/batch while the other factors are not. Consequently, factor E is aliased with the 5-factor interaction ABCDE, and the 2-way interactions, including factor E, are aliased with 4-factor interactions. Meanwhile, the 3-factor interactions that include E are aliased with each other. Hence, we are gaining more details from this experiment than from a regular 1/2 fractional factorial with a defining relation, I = ABCDE.

The results were studied using the Analysis of Variance (ANOVA) technique, see Appendix B. The factor levels and their corresponding numeric values (−1,1) are listed in Table 4. The test results are illustrated in Section 4. Meanwhile, the generated regression models and their performance are illustrated in Section 5.3. The details of the regression model are provided in Appendix C.

Table 3
Main effects, interaction effects and their aliases for the 1/2 fractional factorial DOE.

Main effects	Interactions	
A = *BCD*	***AB*** = *CD*	***ABE*** = *CDE*
B = *ACD*	***AC*** = *BD*	***BCE*** = *ADE*
C = *ABD*	*AD* = ***BC***	***ACE*** = *BDE*
D = *ABC*	***AE*** = *BCDE*	
E = *ABCDE*	***BE*** = *ACDE*	
	CE = *ABDE*	
	DE = *ABCE*	

3. Experimental procedures

3.1. Materials

The material subject to investigation matches the UNS N07718 in API standard 6ACRA, here referred to as nickel alloy 718. Two types of powder have been used in this study; virgin and recycled powders: powder as delivered from the supplier and powder that has been in the build chamber for at least one heat cycle, respectively. When compared extensively, the only significant difference between the powders was in the physical morphology, leading to the physical uniformization of the powders for this study. Table 5 shows the main physical properties of the virgin and recycled powders while Fig. 3 shows SEM images for the powder particles.

3.2. Sieving

Gilson 8in and Retsch AS 200 vibratory sieve shaker systems were used in this study to sieve 279 kg of powder. A schematic representation for the vibratory sieve shaker setup is shown in Fig. 4. Each shaker consists of a tower of four 8 in-mesh stainless steel sieves with sizes of 45, 32, 25, and 20 μm and a pan for collecting fine powder with particle size less than 20 μm. A batch of 500 g was placed on the top sieve and allowed to vibrate with the sieving parameters shown in Table 6. A 10 g sample was taken from each powder cut to measure the PSD using laser diffraction consistent with ASTM B822-17 utilizing a dry cell Horiba L960 particle size analyzer [45].

3.3. Manufacturing

L-PBF experiments were conducted on an SLM 280 machine (SLM® Solutions GmbH) equipped with two IPG ytterbium fiber lasers with a 3D vario scan optics and maximum laser power of 400 W. Parts were printed with a stripe pattern of 10 mm width and a 67° pattern rotation. A layer thickness of 60 μm was used to manufacture all the samples. During the additive manufacturing process, the oxygen content in the manufacturing chamber was kept below 0.1% by allowing the flow of argon gas. During each experiment, 6 tensile bars, 6 Charpy impact

Table 4
Different levels for each factor and their corresponding numeric value.

Factor	Type [A]	Range [B]	D50 [C]	Moisture content [D]	VED [E]
Level 1 (−1)	Virgin	20:75 μm	27 μm	Dry	48.6 J/mm^3
Level 2 (+1)	Recycled	0:75 μm	37 μm	Ambient	36.1 J/mm^3

Table 5
Virgin and recycled powder characteristics.

Physical powder characteristic		Virgin	Recycled	Recycled (Far)	Recycled (Close)
Particle size distribution	D10 [μm]	19.494 ± 0.06	19.602 ± 0.03	19.607 ± 0.21	21.818 ± 0.36
	D50 [μm]	31.766 ± 0.07	30.993 ± 0.09	31.148 ± 0.39	34.389 ± 0.29
	D90 [μm]	52.603 ± 0.06	50.453 ± 0.39	50.220 ± 0.40	54.041 ± 0.22
	Mean [μm]	34.450 ± 0.15	33.490 ± 0.17	34.487 ± 0.22	37.370 ± 0.10
Sphericity	Mean	0.846 ± 0.09	–	0.816 ± 0.10	0.804 ± 0.09
	Median	0.86	–	0.83	0.81
Flowability	Hall [s/50g]	14 ± 0.07	–	16 ± 0.21	16 ± 0.12
Porosity	[% pores]	4.07 ± 0.12	–	4.33 ± 0.16	4.75 ± 0.17

Table 6
Sieving parameters for both virgin and recycled powders.

	Virgin powder		Recycled powder	
	Time (min)	Amplitude /frequency	Time	Amplitude /frequency
Gilson vibratory sieve	50	6	70	6
Retsch vibratory sieve	60	40	60	35

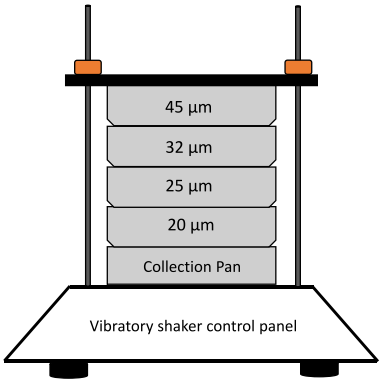


Fig. 4. Schematic representation for the vibratory sieving setup used in the study.

specimens, and 6 cubes for density measurements were printed. Half of those specimens were fabricated using VED 1, while the other half used VED 2. Additionally, 6 powder capsules were located on each side (left and right) of the substrate relative to the gas flow. Fig. 5 shows the layout of all the fabricated specimens and the powder capsules.

After the manufacturing process, the samples were cut from the substrate with wire electrical discharge machining (EDM) from the as-built state to near specification. Then it was subjected to solutionizing and single-age heat treatment and brought into tolerance with traditional machining.

3.4. Mechanical testing

Tensile testing was completed on an MTS Insight 30 kN electromechanical test system. ASTM Standard E8M-16a [46] round specimen no. 4 with type 1 machined was chosen for a combination of size and measurement compatibility with the load cell. Three sample specimens for each condition were built in semi-cylindrical geometries to minimize printed material and ensure full support during the AM process. These parts were cut into cylinders via wire EDM and finally turned down to tolerance. The samples were threaded into threaded H13 inserts gripped by the load cell to minimize outside forces affecting the tension test. A laser extensometer was used instead of a mechanical extensometer because of the high elongation of the material and the increased gauge length section in E8M samples compared to E8. The specimens were tested using control method C (7.6.4.3) at a cross-head speed of 0.015 mm/min. The load versus extension was recorded, and the elastic modulus (E), 0.02% yield stress (σ_y), yield strain (ϵ_y), ultimate tensile strength (UTS) (σ_{UTS}), strain at UTS (ϵ_{UTS}), elongation at fracture, resilience (J/m^3), and toughness (J/m^3) were calculated for each sample. Resilience and toughness were calculated by integrating under the stress–strain curve in the elastic and plastic regions, respectively. Integration was done using the trapezoidal rule numerical integration method.

Charpy impact testing was conducted using an Instrom 450 MPX pendulum impact tester in accordance with E23-18 [47] with v-notched specimens at room temperature (21 °C). The specimens had 55 mm, 10 mm, and 10 mm length, width, and thickness, respectively. The v-notch was located at the center of the specimen length with an angle of 45°. Impact energy absorbed was recorded for each sample, as well as qualitative data on fracture surfaces. For each condition, three Charpy and tensile samples were tested.

3.5. Density measurement

Both density measurements on the final built products and the powder bed at the time of build were determined, the latter being an in-situ powder measurement in contrast to characterization techniques like tapped density. The powder bed quality is proposed to significantly impact the final part outcome [48,49]. Quality is defined as a combination of uniformity, predictability, and consistency of the powder bed across locations and builds. Similar processes to measure the powder bed have been proposed in the literature with varying levels of success [50]. The in-situ powder bed density presented herein was accomplished with a technique similar to Jacob et al. [50] but with a modified container design. The container is fabricated during the layer-by-layer process. Once removed from the build plate, it is unsealed, emptied, and refilled with a known liquid to determine the internal volume. The powder is then weighed to determine its density.

Fabricated cube of 15 × 15 × 15 mm^3 were machined into a vertical and horizontal cross-section using wire EDM. After mounting in Allied red phenol, samples were polished to 0.04 μm. After polishing, the parts were imaged on a DXS 500 optical microscope by Olympus America, which includes an auto-stitch function that allows capturing images as large as 12 × 6 mm^2 with the smallest measurable pores being below 20 μm. Using image analysis software, the observable pore area was recorded as a percent of the total observable area. This optical microscope density is recorded as less than the ratio of pore area to total area. The challenge associated with this measurement technique is that the cross-section of material will not accurately capture the actual

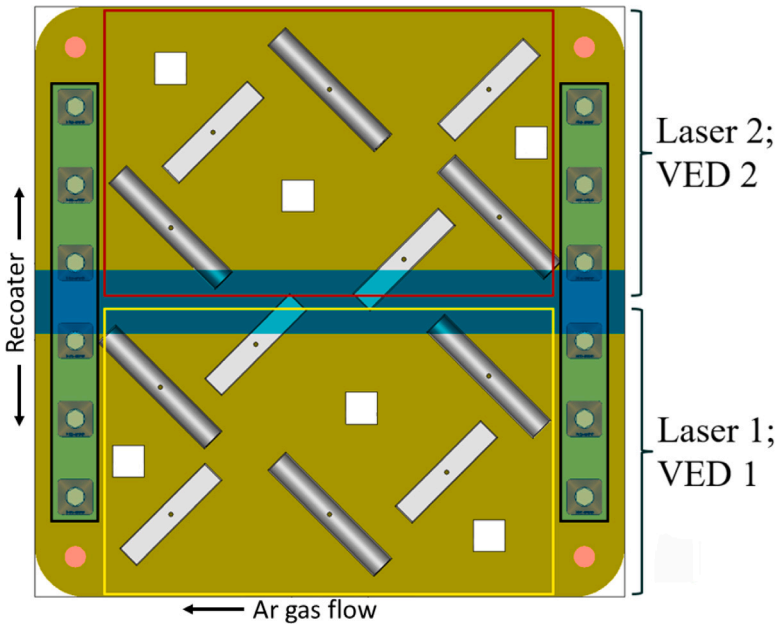


Fig. 5. CAM layout of the substrate showing the different manufactured specimens.

diameter of pores, under-representing the size of pores since the mid-plane of spherical pores will not lie on one plane. This misestimation was the impetus for large sampling areas to increase the statistical likelihood of accurately recording the area porosity representative of true bulk porosity.

4. Results and discussion

4.1. Optical microscopy densification results

The printed cubes' optical Microscopy (OM) relative density was measured for each experimental run as shown in Table 7. The results are also illustrated in the bar chart in Fig. 6. It can be observed from Fig. 6 that the VED factor has a high impact on the relative density. Besides, the average OM densification values of VED 1 and VED 2 are 99.88% and 98.77% for the virgin powder and 99.56% and 98.96% for recycled powder, respectively. This indicates that VED 1 yields better densification for virgin powder than recycled powder. On the other hand, VED 2 yields better densification for recycled powder compared to virgin powder. Moreover, the average relative density of both powder types for VED 1 was higher than that for VED 2, with averages of 99.72% and 98.86%, respectively.

4.2. Density of powder capsules

The powder-bed packing density can be tested via powder density capsules that contain powder collected during the printing process. For each powder batch, 12 samples were collected, of which six were located at the entrance and six at the exit of the gas flow, see Fig. 7. As this process does not include the sintering of the powder, the VED level does not impact the results. However, a different factor that might affect the packing density is the location of the powder with respect to the gas flow inside the chamber. Hence, factor E, in this case, represents the location parameter. As there are two possible locations, namely, gas flow entrance and exit, the experimental design remains the same for the remaining tests.

On average, the recycled powder seems to perform better (58.09% vs. 56.63%) than the virgin powder (see Fig. 8). Moreover, the recycled powder at the gas flow exit seems to have a higher power density, while the virgin powder at the gas flow entrance had a higher powder density on average. This means that the interaction effect between the powder type and sample location in relation to the gas flow is significant, see Table 8.

Table 7
OM relative density results for the additively manufactured cubes. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

Exp.	A	B	C	D	E	OM densification [%]
1	−1	−1	−1	−1	1	98.19
2	−1	−1	1	1	1	98.92
3	−1	1	−1	1	−1	99.98
4	−1	1	1	−1	−1	99.98
5	1	−1	−1	1	1	98.46
6	1	−1	1	−1	1	99.27
7	1	1	−1	−1	−1	99.63
8	1	1	1	1	−1	99.63
9	−1	−1	−1	−1	−1	99.59
10	−1	−1	1	1	−1	99.96
11	−1	1	−1	1	1	98.72
12	−1	1	1	−1	1	99.24
13	1	−1	−1	1	−1	99.35
14	1	−1	1	−1	−1	99.62
15	1	1	−1	−1	1	99.18
16	1	1	1	1	1	98.91

4.3. Tension test

Figs. 9 and 10 show the stress–strain curves for nickel alloy 718 tensile specimens manufactured by the L-PBF process under the 16 different conditions as listed in Table 2. Fig. 9 shows the 16 stress–strain curves overlapped to study the effect of both VED levels on the mechanical properties. It was observed that specimens manufactured using VED 1 have higher UTS, yield strength, toughness, and elongation at fracture. It also shows that VED 1 samples undergo plastic deformation and necking and then fracture afterward. On the other hand, samples manufactured using VED 2 show plastic deformation until the UTS point, then fracture suddenly. This can be attributed to the formation of higher porosity in samples manufactured using VED 2.

In Fig. 10, the plots for all the other factors (factors A, B, C, and D) were split into two plots. Each plot represents a different VED level

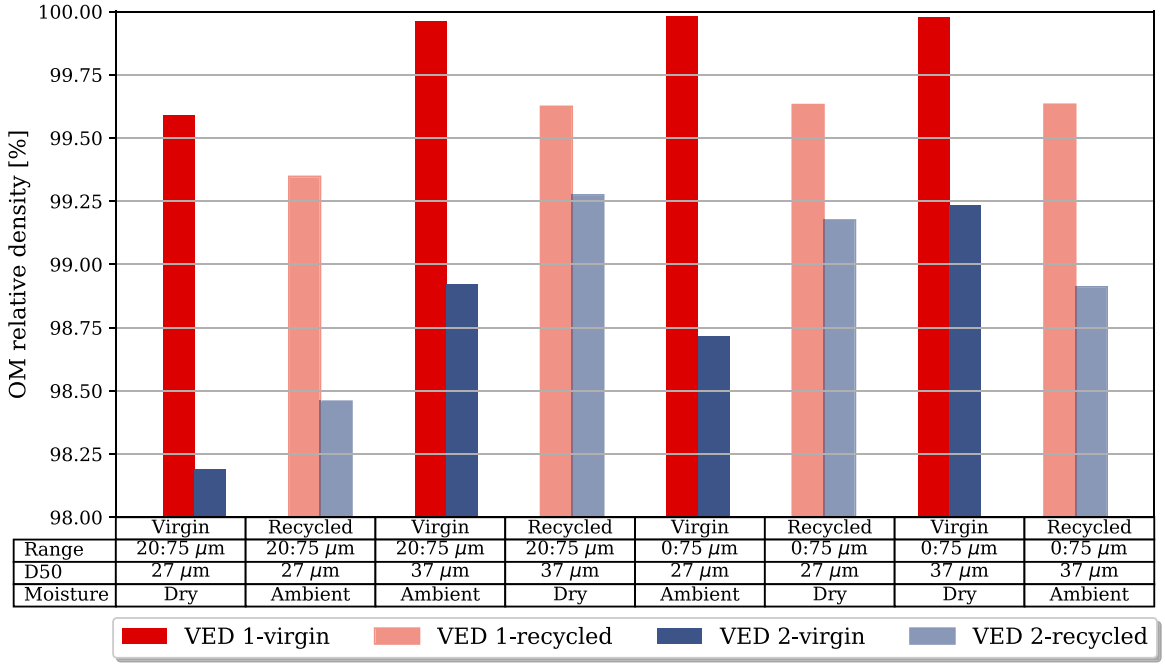


Fig. 6. Bar chart for the OM relative density results.

Table 8
Powder-bed packing density results of powder capsules. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: location w.r.t. the gas flow.

Exp.	A	B	C	D	E	Powder density [%]
1	−1	−1	−1	−1	1	58.00
2	−1	−1	−1	−1	−1	58.14
3	−1	−1	1	1	1	55.61
4	−1	−1	1	1	−1	56.53
5	−1	1	−1	1	1	56.39
6	−1	1	−1	1	−1	57.17
7	−1	1	1	−1	1	56.03
8	−1	1	1	−1	−1	55.17
9	1	−1	−1	1	1	58.23
10	1	−1	−1	1	−1	56.99
11	1	−1	1	−1	1	58.43
12	1	−1	1	−1	−1	56.63
13	1	1	−1	−1	1	59.06
14	1	1	−1	−1	−1	58.34
15	1	1	1	1	1	58.72
16	1	1	1	1	−1	58.38

since there is an extreme variability between mechanical properties for the specimen manufactured using VED 1 and VED 2 levels. From Fig. 10(a), the virgin powder seems to outperform the recycled powder regardless of the VED level used. Meanwhile, when plotting the stress–strain curves for the two particle size ranges and the two D50 values, as shown in Figs. 10(b) and 10(c), no visual inference of the effect of any of the PSD factors can be obtained. Besides, when considering both factors B and C together in Fig. 10(d), no inferences could be obtained. Similarly, from Fig. 10(e) no remarkable difference was observed for the effect of the moisture content level on the mechanical properties. Hence, a more in-depth statistical analysis is needed to determine the significance of these factors. Moreover, Table 9 summarizes the mechanical properties calculated from the test.

4.4. Charpy impact test

The high energy fracture caused by a Charpy pendulum causes rapid propagation of cracks and a complete overload of samples. This test

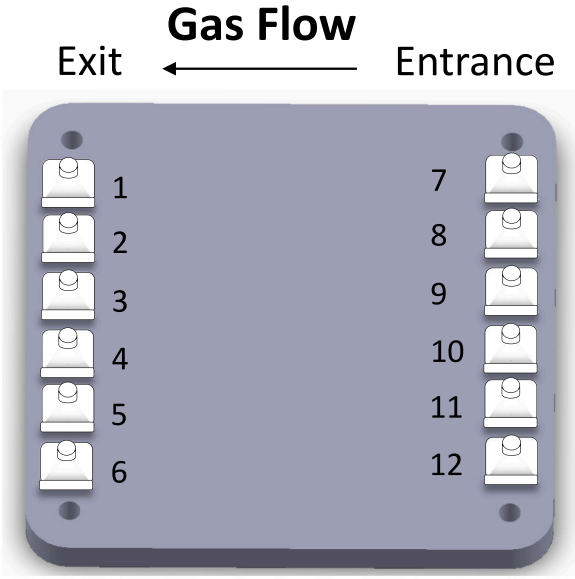
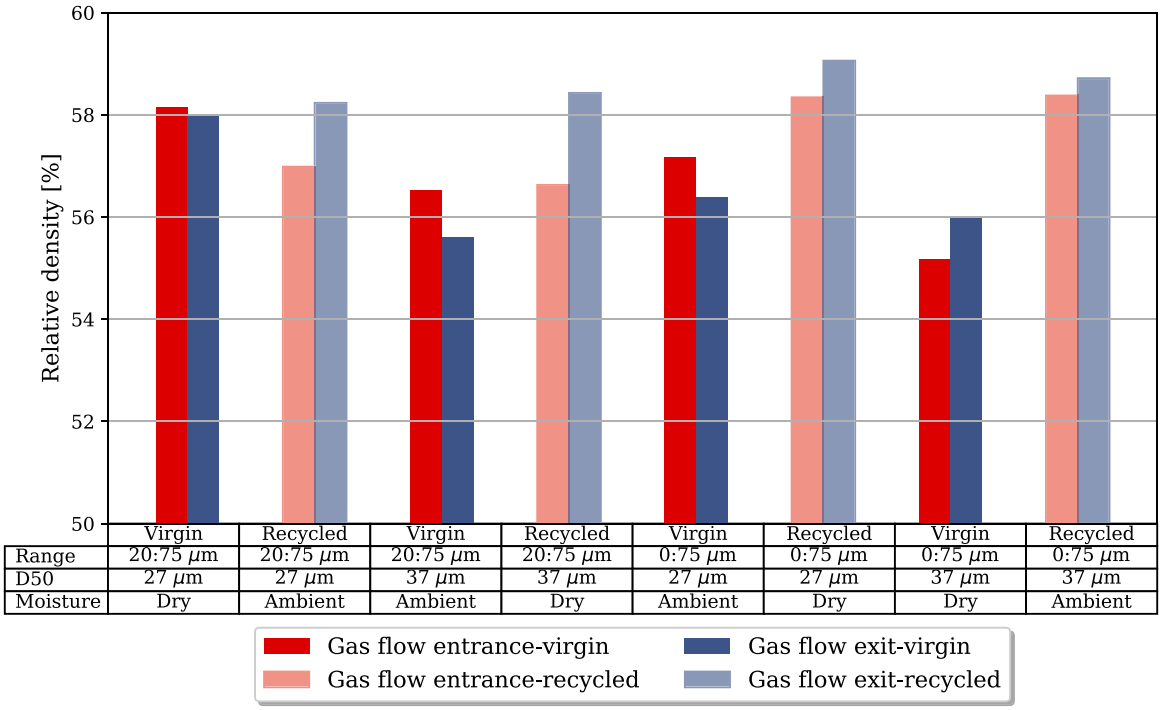


Fig. 7. Capsule location on the substrate with respect to the gas flow direction.

was intentionally chosen because the rapid nature of the fracture, in theory, prevents work hardening and should increase the differentiation between high-porosity and low-porosity samples in a quantifiable mechanical test. The impact energy values measured from the Charpy test for the 16 experiments are shown in Table 10 and are presented in a bar chart in Fig. 11. Like the previous test results, the VED factor has the most significant effect on the impact energy, with averages of 56.94 J and 23.46 J for VED 1 and VED 2, respectively. Considering VED 1 samples only, samples that share the same PSD (particle size range and D50), represented by each two adjacent red and light red bars in Fig. 11, are showing better performance for the dry powder vs. the ambient one regardless of the powder type. In addition, the 20:75 μm range powder with similar PSD values has the highest impact strength



	Virgin	Recycled	Virgin	Recycled	Virgin	Recycled	Virgin	Recycled
Range	20:75 μm	20:75 μm	20:75 μm	20:75 μm	0:75 μm	0:75 μm	0:75 μm	0:75 μm
D50	27 μm	27 μm	37 μm	37 μm	27 μm	27 μm	37 μm	37 μm
Moisture	Dry	Ambient	Ambient	Dry	Ambient	Dry	Dry	Ambient

Fig. 8. Powder-bed packing density of nickel alloy 718 using powder capsules.

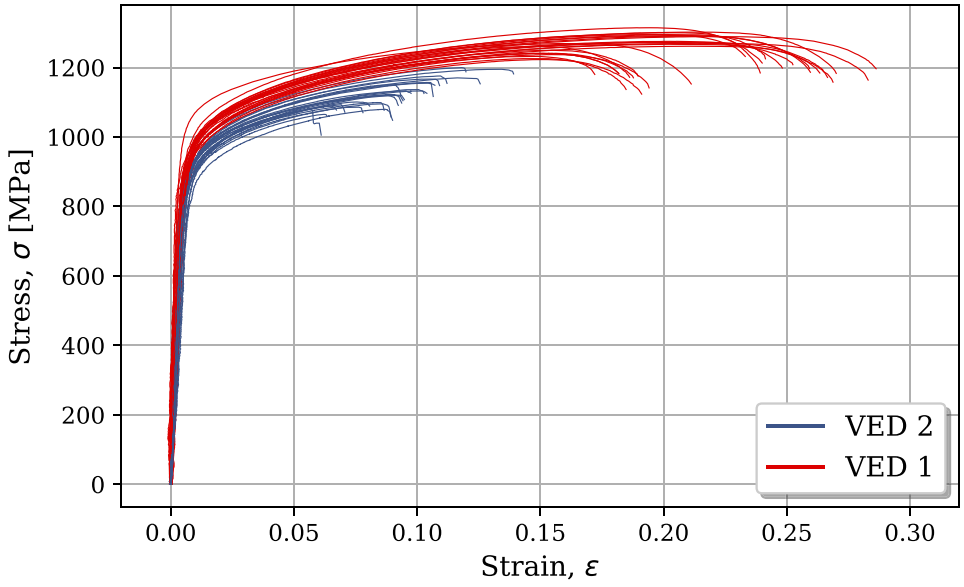


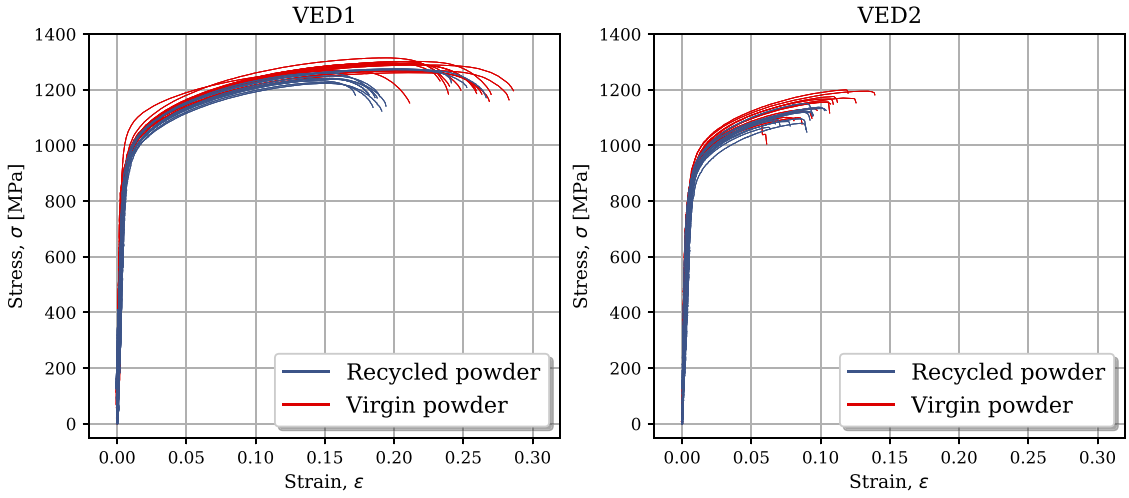
Fig. 9. Effect of VED levels (Factor E) on the mechanical properties of nickel alloy 718 samples manufactured by L-PBF.

irrespective of its type. On the other hand, for VED 2 samples, the virgin powder systematically outperforms the recycled powder for every two samples with the same PSD, regardless of the moisture content level.

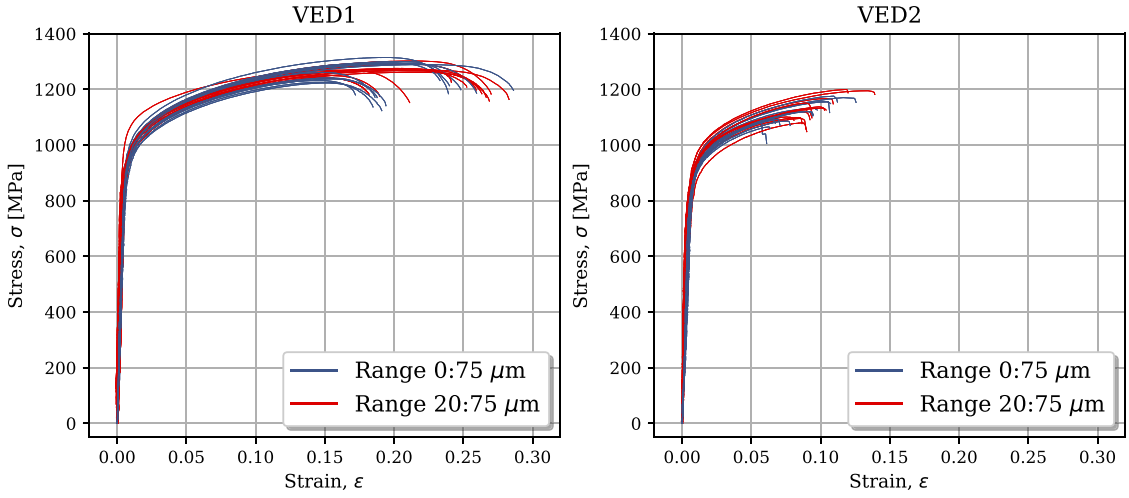
5. Statistical analysis

5.1. Analysis of variance (ANOVA)

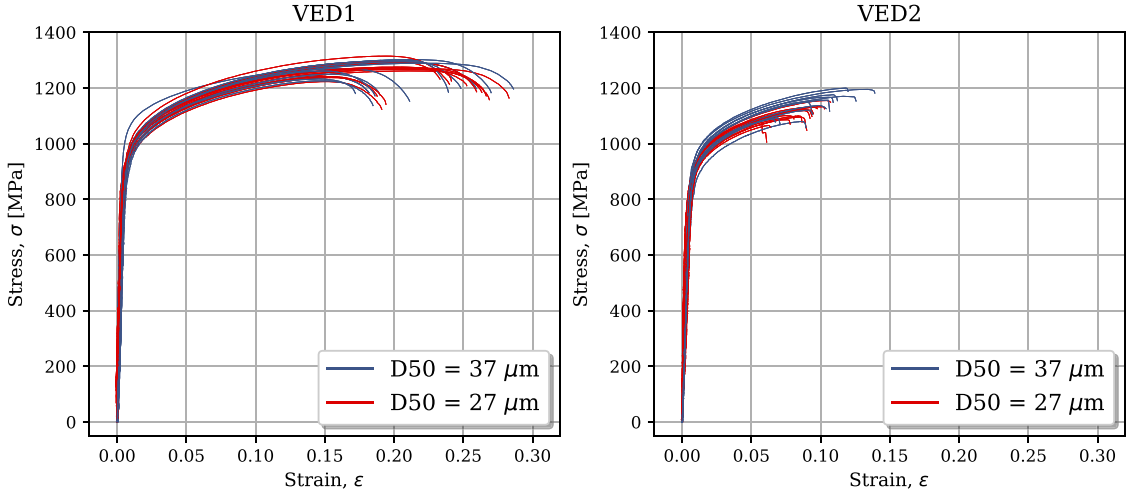
The ANOVA results for the different QoIs are summarized in Table 11. The significant factors/interactions are determined based on their p-value being less than 0.05 (level of significance, α). Some factors/interactions are included that had a p-value slightly higher than 0.05, indicated with an asterisk '*', these are weakly significant factors/interactions. The significant factors are shown in Table 11 with their corresponding F-values. The higher F-values indicate a higher variability due to this factor/interaction. Moreover, the factor with the highest F-value is highlighted in bold case. From the ANOVA results, factors A (powder type), B (PSD range), and E (VED) are the most commonly statistically significant factors. Meanwhile, interactions BC (PSD range and D50) and BCE are the most commonly statistically significant two-factor and three-factor interactions. It should be mentioned that the combination of factors B and C with their interaction BC represent the combined effect of the PSD factor. Factor D appears to be the least significant factor, being only significant with respect to the Charpy impact energy and ϵ_{UTS} (which is one of the least important tensile test QoIs). Moreover, the interaction DE is the least significant two-factor interaction.



(a)

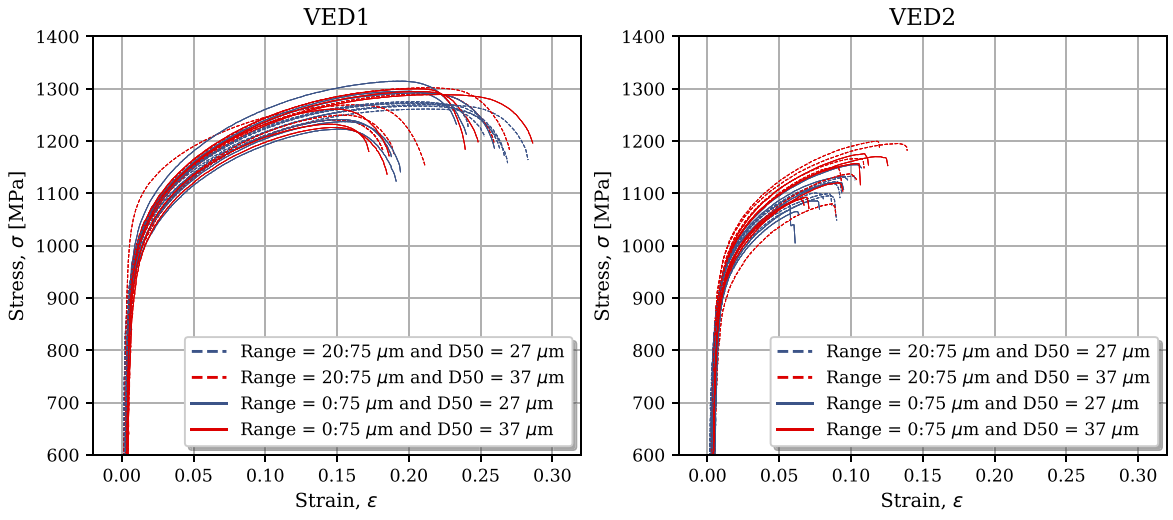


(b)

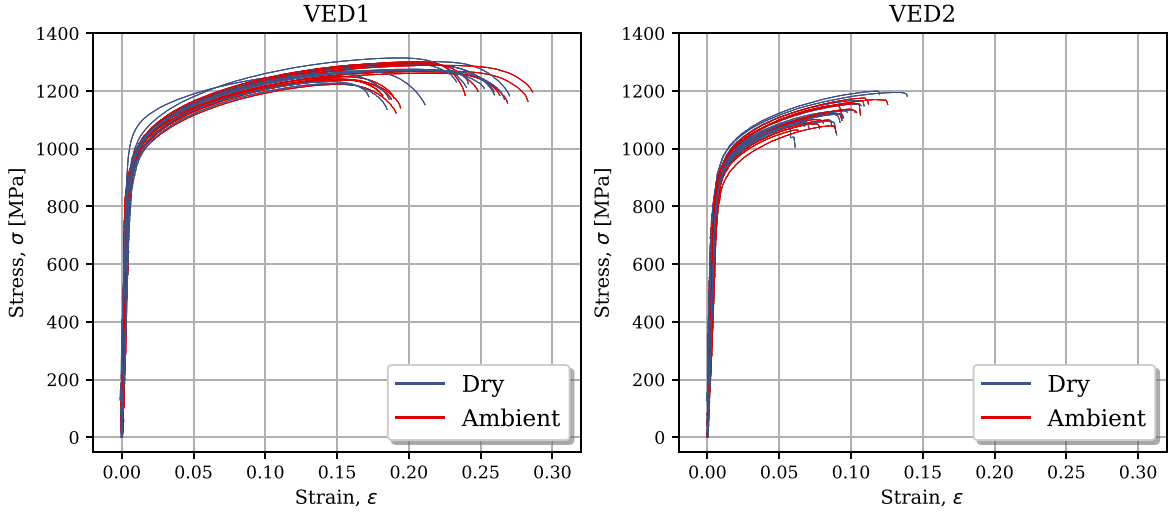


(c)

Fig. 10. Effect of (a) powder type (factor A), (b) particle size range (factor B), (c) median particle size; D50, (factor C), (d) PSD represented by both factors B and C, (e) moisture content (factor E), on the mechanical properties of nickel alloy 718 specimens manufactured by L-PBF.



(d)



(e)

Fig. 10. (*continued*).

Table 9
Mechanical properties of nickel alloy 718 L-PBF tensile specimens for the 16 experimental runs. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

Exp.	A	B	C	D	E	E [GPa]	σ_y [MPa]	ϵ_y [10^{-3} m/m]	σ_{UTS} [MPa]	ϵ_{UTS} [m/m]	Frac. elongation [%]	Resilience [kJ/m^3]	Toughness [kJ/m^3]
1	−1	−1	−1	−1	1	215.62	923.74	6.781	1169.29	0.083	8.71	3.14	88.17
2	−1	−1	1	1	1	200.27	890.59	6.510	1187.22	0.118	12.23	2.90	133.46
3	−1	1	−1	1	−1	190.12	931.81	6.954	1300.85	0.200	24.43	3.24	295.67
4	−1	1	1	−1	−1	188.05	930.54	6.988	1294.65	0.207	25.79	3.25	311.50
5	1	−1	−1	1	1	230.99	896.04	6.121	1178.06	0.092	9.51	2.74	97.18
6	1	−1	1	−1	1	167.72	890.06	7.357	1163.87	0.092	9.56	3.28	96.28
7	1	1	−1	−1	−1	172.22	978.27	8.037	1302.70	0.153	19.10	3.93	217.79
8	1	1	1	1	−1	169.12	964.96	7.710	1309.07	0.148	18.10	3.72	206.80
9	−1	−1	−1	−1	−1	307.61	963.09	5.410	1336.14	0.214	27.59	2.61	329.11
10	−1	−1	1	1	−1	182.31	976.62	7.363	1287.24	0.195	24.69	3.60	298.31
11	−1	1	−1	1	1	170.85	869.80	7.144	1144.02	0.099	10.24	3.11	106.49
12	−1	1	1	−1	1	153.91	891.52	7.804	1167.72	0.110	11.48	3.48	121.74
13	1	−1	−1	1	−1	210.07	978.27	6.917	1344.67	0.201	25.35	3.39	303.26
14	1	−1	1	−1	−1	146.29	831.46	7.800	1314.65	0.155	18.97	3.19	215.64
15	1	1	−1	−1	1	144.40	919.97	8.640	1151.51	0.075	7.88	3.99	76.55
16	1	1	1	1	1	153.94	915.79	7.910	1161.99	0.075	7.74	3.62	75.73

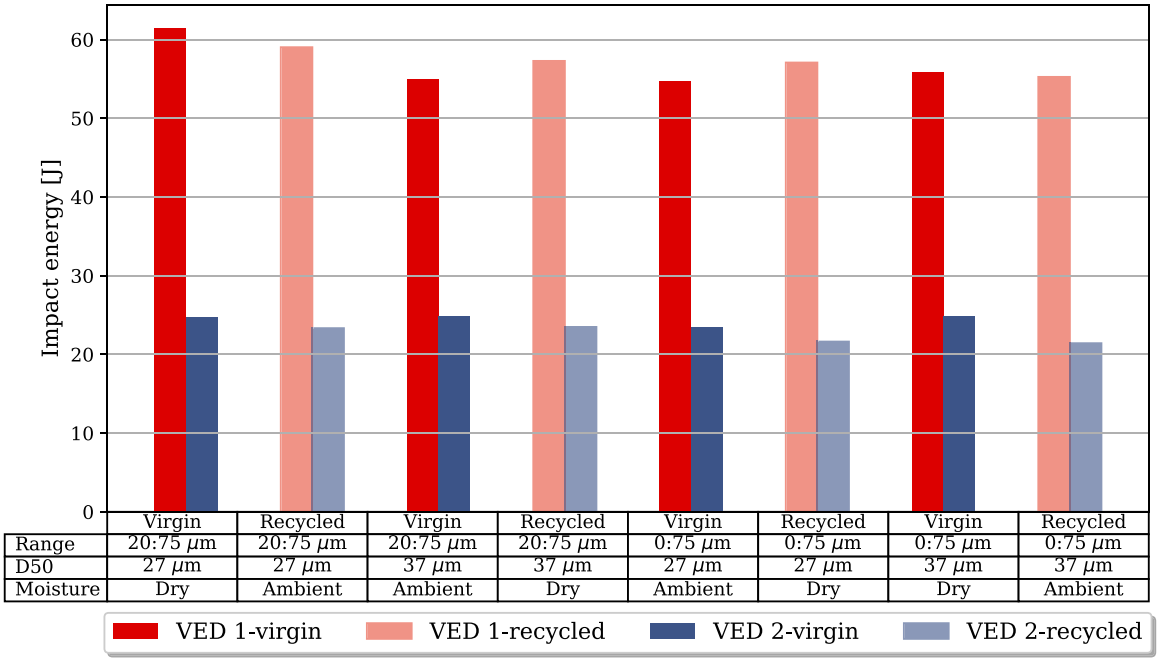


Fig. 11. Charpy impact test results for nickel alloy 718 samples manufactured by L-PBF.

Table 10
Impact energy values obtained from Charpy impact test. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

Exp.	A	B	C	D	E	Impact energy [J]
1	−1	−1	−1	−1	1	24.70
2	−1	−1	1	1	1	24.89
3	−1	1	−1	1	−1	54.74
4	−1	1	1	−1	−1	55.84
5	1	−1	−1	1	1	23.33
6	1	−1	1	−1	1	23.49
7	1	1	−1	−1	−1	57.08
8	1	1	1	1	−1	55.25
9	−1	−1	−1	−1	−1	61.38
10	−1	−1	1	1	−1	54.89
11	−1	1	−1	1	1	23.45
12	−1	1	1	−1	1	24.80
13	1	−1	−1	1	−1	59.02
14	1	−1	1	−1	−1	57.28
15	1	1	−1	−1	1	21.63
16	1	1	1	1	1	21.42

Because of the high variability between the VED levels, the ANOVA testing is repeated using the samples from VED 1 and VED 2 separately. Thus, using VED as a blocking factor to eliminate its effect and allow the other factors to show whether they are significant or not. The ANOVA results when considering VED 1 only and VED 2 only are shown in Tables 12 and 13, respectively. The factors/interactions that were significant in the combined ANOVA and are no longer significant are signaled by a '↓' next to them. While the factors/interactions that were not significant and became significant are signaled by a '↑' next to them.

When only considering VED 1 samples in ANOVA, there became no significant factors for the modulus of elasticity QoI, as opposed to when using all VED samples. Factor C, the D50, is shown to be significant, especially with the QoIs collected in the plastic region (UTS, ϵ_{UTS}, toughness, and elongation at fracture). Factor A, powder type, became significant with respect to the UTS and the OM densification. Other than that, the ANOVA results seem to be consistent with that of the general case with all VED levels.

Meanwhile, when only considering VED 2 in the ANOVA, many factors/interactions that were significant no longer became significant. As F-value in ANOVA represents the ratio of the factor variance/variability to the error variance (noise), the higher the F-value the more significant a factor is. A low F-value means that either the factor is less significant or that the noise is high. Thus, having many factors/interactions become insignificant indicates a higher noise, unobserved variability, associated with the experiment. This could be due to the high porosity with VED 2. Again, as with VED 1 only, factor C became more significant, while interaction BC became insignificant for all QoIs.

5.2. Effect plots

The main and the interaction effect plots are used to provide a deeper understanding of the effect of the different factor levels, rather than the variability significance, on the QoIs. For the entire effect plots of all QoIs, please refer to Figs. D.1–D.10 in Appendix D. One major observation from the effect plots was that of factor B (PSD range). The expectation was that with the inclusion of the finer particles in the case of factor level B2, the part densification would increase due to the presence of interstitial particles. This was indeed observed in the increase of the OM densification and powder-bed packing density results. However, the main QoIs of the tension test, as well as the Charpy impact energy test, showed a reversed observation with the 20:75 μm powder range (factor level B1) showing better results than the 0:75 μm powder range (factor level B2). These observations are shown in Fig. 12, which includes the main effect plots of factor B for the main QoIs. There are a number of possible explanations for these outcomes, though there is little literature on the result currently. Classic sentiment in AM aligns with simplicity, increasing powder-bed packing density, increasing part density, increases mechanical performance and part outcomes. It is possible that the difference here is due to increased powder uniformity by removing interstitial particles that can aggregate in certain locations in the build volume, or due to a difference in energy required to generate a melt pool or melt pool instability caused by vaporization of small particles. Regardless of the cause, the outcome is exciting, especially considering the possibility of removing extremely small particles improves health, safety, and environmental considerations to handling metal powders.

Again, to eliminate the effect of the high porosity in samples manufactured using VED 2, hence eliminating the associated noise with VED 2 and confirming our aforementioned finding about factor B, the same

Table 11
ANOVA results for the different QoIs. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

Factor/ interaction	E	Resilience	σ_y	ϵ_y	σ_{UTS}	ϵ_{UTS}	Elong. at fracture	Toughness	Impact energy	Powder density[a]	OM density
A		6.84	4.54	8.98		98.54	77.59	99.24	5.53	**38.51**	
B	**6.32**	**10.56**		**11.65**	15.25	11.7	12.45	13.55	31.63		6.43
C	5.41			3.89*					8.07	13.11	10.73
D						2.93*			12.28		
E			**79.06**		**933**	**959.9**	**1,075.66**	**1,236.07**	**10,403.77**		**84.6**
AB			11.72			17.04	8.93	10.46		16.88	
AC						14.79	11.73	14.81		9.75	
BC	2.92*			4.26	8.3	3.03*	4.35	5.09	10.01		6.78
AE						11.97	15.1	21.6	13.8	7.25[a]	7.38
BE	4.3						3.53	3.67*	3		
CE					9.42	21.51	16.54	18.28	15.81		
DE		6.64		4.22					5.49		
ABE											
BCE						14.85	13.05	14.33	6.54		
ACE					4.10*						

[a] Factor E for the powder density QoI represents the location w.r.t. the gas flow factor rather than the VED factor.

Table 12
ANOVA results for the different QoIs considering VED 1 only. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

Factor/ interaction	E	Resilience	σ_y	ϵ_y	σ_{UTS}	ϵ_{UTS}	Elongation at fracture	Toughness	Impact energy	OM density
A		6.84	4.54	8.98		98.54	77.59	99.24	5.53↓	
B	6.32↓	10.56		11.65	15.25	11.7	12.45	13.55	31.63	6.43
C	5.41↓			3.89*					8.07	10.73
D						2.93*↓			12.28	
AB			11.72			17.04	8.93	10.46		
AC						14.79	11.73	14.81		
BC	2.92*↓			4.26	8.3	3.03*	4.35	5.09	10.01	6.78
VED 1 Only										
A		5.23	3.54*	10.02	8.04↑	82.58	58.05	74.84		18.44↑
B		4.35*		3.86*	17.4	10.09	10.53	10.99	38.21	5.57
C				4.98	18.96↑	12.60↑	9.91↑	10.79↑	32.83	4.73
D									24.15	
AB			4.82			10.42	4.10*	5.69		
AC						6.27	5.37	7.16		
BC				7.22	19.12	14.42	11.7	12.8	23.11	4.81

Table 13
ANOVA results for the different QoIs considering VED 2 only. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

Factor/ interaction	E	Resilience	σ_y	ϵ_y	σ_{UTS}	ϵ_{UTS}	Elongation at fracture	Toughness	Impact energy	OM density
A		6.84↓	4.54↓	8.98↓		98.54	77.59	99.24	5.53	
B	6.32	10.56		11.65↓	15.25	11.70	12.45↓	13.55↓	31.63	6.43
C	5.41↓			3.89*↓					8.07↓	10.73
D						2.93*↓			12.28↓	
AB			11.72			17.04↓	8.93	10.46		
AC						14.79	11.73	14.81		
BC	2.92*↓			4.26↓	8.30↓	3.03*↓	4.35↓	5.09↓	10.01↓	6.78↓
VED 2 Only										
A						8.40	19.80	24.59	14.24	
B	9.76	6.48			4.75	3.45*			5.86	3.92*
C						8.39↑	6.82↑	7.85↑		7.93
D		4.00*↑								
AB			7.17				5.55	5.18		
AC						8.24	7.30	8.80		
BC										

plots were generated for the data using VED 1 and VED 2 samples only. The results, shown in Figs. 13 and 14, respectively, show the same trend when using all VED samples.

5.3. Regression models

It is important to be able to predict each quantity of interest based on the factors and interactions. Only the significant factors/interaction terms are considered in the predictive models to prevent over-fitting. It is also a general practice that the factors that constitute significant interactions are included in the model even if they are insignificant [51].

To verify the aforementioned approach, the performance of predictive models that include all the terms (full model) and that include the significant terms only (reduced model) is tested using the mean absolute percentage error (MAPE) metric:

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i^{obs} - y_i^{pred}|}{y_i^{obs}} * 100\%$$

where, N is the number of experiments, y^{obs} is the observed value and y^{pred} is the predicted value from the model. Moreover, the error values were reported for the leave-one-out cross-validation (LOOCV)

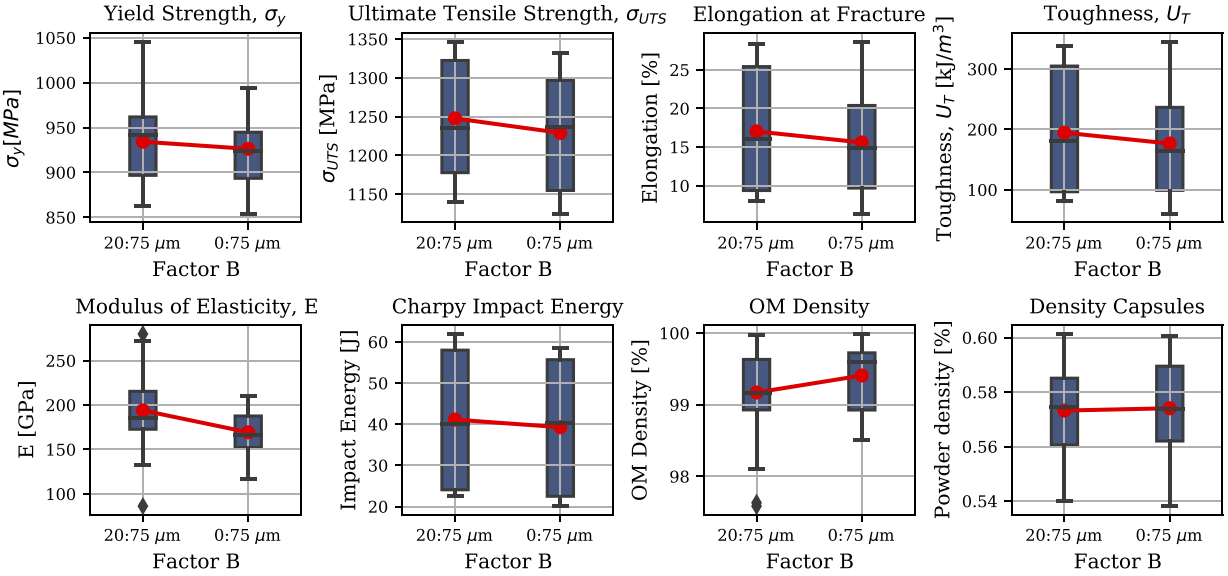


Fig. 12. Collection of the effect plots of factor B on the main QoIs.

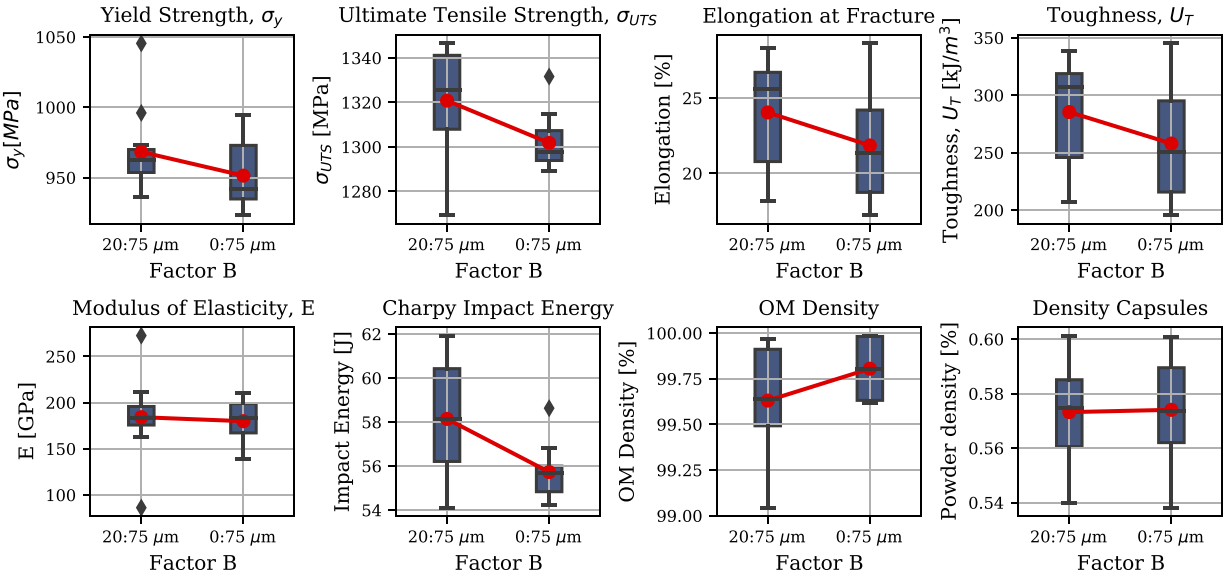


Fig. 13. Collection of the effect plots of factor B on the main QoIs by using VED 1 test results only.

scheme as a surrogate for the testing error values. This is because of the small amount of collected data. The LOOCV MAPE results are shown in Table 14. For all the models, the MAPE values are reasonable. The reduced models, including the significant terms only, are performing better in all cases than these in the full model. The relatively higher MAPE values for the modulus of elasticity, resilience, and yield strain can be due to the approximate methods used for their estimation. A summary of the regression coefficients is given in Table 15 for all the QoIs.

6. Conclusion

This paper assesses the effects of the different powder characteristics and VED on the densification and mechanical properties (excluding fatigue and corrosion) of nickel alloy 718 parts manufactured by L-PBF. The study is done using a statistically designed experiment (DOE) that allows the assessment of these powder characteristics as well as most of their interactions. The suggested DOE can assess all five main effects as well as ten of the interaction effects. The DOE is a resolution IV design, i.e., it has clear main effects, and the two-factor interactions are aliased with each other (thus, half of them can be assessed). Designing this DOE without reducing factor E allows the analysis of 15 main/interaction effects with only eight builds/batches of powder.

Tension tests, Charpy impact test, optical microscopy densification, and powder-bed packing density tests were performed. The QoIs of each test are analyzed using the analysis of variance (ANOVA) technique, and based on these analyses, a regression model is built that uses the significant factors/interactions only as covariates.

VED, powder type, and particle size range are the most commonly statistically significant factors, while the VED level followed by the powder type is the most significant factor with respect to the tensile strength. Meanwhile, the interaction BC (PSD range and median), and interaction BCE (PSD range, PSD median, and the VED) are the most common significant 2-factor and 3-factor interactions, being significant to eight and four of the QoIs, respectively.

The moisture content had almost no significant effect on the tensile test results as well as the OM and powder-bed packing densification

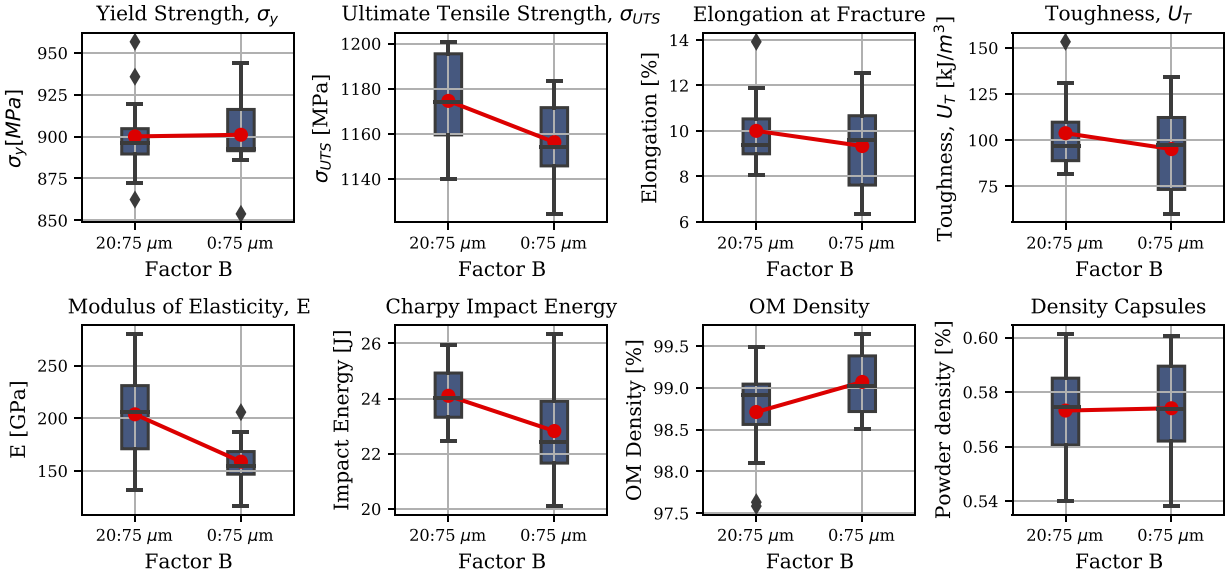


Fig. 14. Collection of the effect plots of factor B on the main QoIs by using VED 2 test results only.

Table 14
Summary of the LOOCV MAPE values for the regression models of each QoI for both the full and reduced models.

Model	E [GPa]	Resilience [kJ/m^3]	σ_y [MPa]	ϵ_y [m/m]	σ_{UTS} [MPa]	ϵ_{UTS} [m/m]	Elong. at frac. [%]	Toughness [kJ/m^3]	Impact energy [J]	$\rho_{cap.}$ [%]	ρ_{OM} [%]
Full	17.8	16.43	2.03	12.51	1.3	7.56	9.25	10.27	3.82	1.83	0.25
Reduced	14.7	14.1	1.83	10.7	1.17	7.24	8.48	9.61	3.51	1.77	0.23

Table 15
Summary of the regression coefficients for the reduced models of each QoI. A: powder type, B: particle size range, C: median particle size, D: moisture content, and E: volumetric energy density.

QoI	1	A	B	C	D	E	AB	AC	BC	AE	BE	CE	DE	ABE	BCE	ACE
E	181.6		−12.32	−11.39		−0.43			8.37		−10.17					
Resilience	3.28	0.2	0.25		−0.01	−0.01							−0.2			
σ_y	930.23	7.1	−3.98			−29.62	11.4									
ϵ_y (10^{-4}*)	71.78	3.84	4.37	2.53	−1.33	0.72			−2.64				−2.63			
σ_{UTS}	1238.39	2.43	−9.32	−2.58		−72.93			6.88			7.33				−4.83
ϵ_{UTS} (10^{-3}*)	138.34	−14.54	−5.01	−0.86	2.51	−45.39	−6.05	−5.63	2.55	5.07		6.8			−5.65	
Elong. at frac.	16.31	−1.78	−0.78	−0.24		−6.64	−0.61	−0.69	0.42	0.79	0.38	0.82			−0.73	
Toughness	185.55	−24.4	−9.02	−3.12		−86.1	−7.92	−9.42	5.53	11.38	4.69	10.47			−9.27	
Impact energy	40.09	−0.5	−1.03	−0.58	−0.69	−16.63			0.41	−0.5	0.39	0.76	0.49		−0.31	
Powder density	57.36	0.69	0.01	−0.47		−0.24[a]	0.44	0.33		−0.36[a]						
OM density	99.3	−0.02	0.13	0.16	−0.44				−0.11	0.12						

[a] Factor E for the powder density QoI represents the location w.r.t. the gas flow factor rather than the VED factor.

tests. On the other hand, it was the third most significant factor, after VED and PSD range, on the Charpy impact energy test. While the moisture content did not have a significant effect on most of the mechanical properties, the flowability of the powder in the machine was severely impaired and required manual intervention before printing.

Including the fine particle sizes of less than 20 μm (PSD range-B2 0:75 μm) provided higher OM density and powder-bed packing densification results. On the other hand, excluding these finer particle sizes (PSD range-B1 20:75 μm) produced more favorable values for the remaining QoIs compared to B2. This means that if the desired QoI, e.g., UTS, is different from a higher-density part, there is no need to specify to the manufacturer the inclusion of finer particles in powder preparation. These finer particles are much more expensive to produce than the coarser ones and have further environmental and health issues.

For the powder-bed packing density, for which the VED has no effect, the fifth factor to investigate was the location of the test capsule with respect to the gas flow. Powder-bed packing density is found to be affected only by the powder type, the D50, and the interaction between the powder type, PSD range, and D50, as well as the interaction between the powder type and the location with respect to the gas flow.

The Charpy impact energy was affected by all the factors and most of the interactions except for AB, AC, ABE, and ACE. It was the only main QoI for which the moisture content is statistically significant.

As there was an extreme difference between most of the test results using VED 1 and VED 2, as well as the increased porosity in the VED 2 samples, the ANOVA analysis was repeated for VED 1 and VED 2 samples separately. Thus, using VED as a blocking factor eliminates its effect and allows the other factors to stand out.

For both cases, factor C (D50) is shown to be generally more significant, especially with the plastic region QoIs, e.g., UTS and elongation at fracture. Moreover, the 20:75 μm range (B1) still shows higher QoI values than 0:75 μm (B2) except for the OM densification and powder-bed packing density.

When using VED 1 only, no factors/interactions are significant with respect to the modulus of elasticity. Other than that, the ANOVA results are generally consistent with that for all VEDs.

On the other hand, when VED 2 only is used, the PSD range is the only significant factor/interaction with respect to the modulus of elasticity. Moreover, many factors/interactions that were significant when considering all VEDs are not significant. This can be interpreted

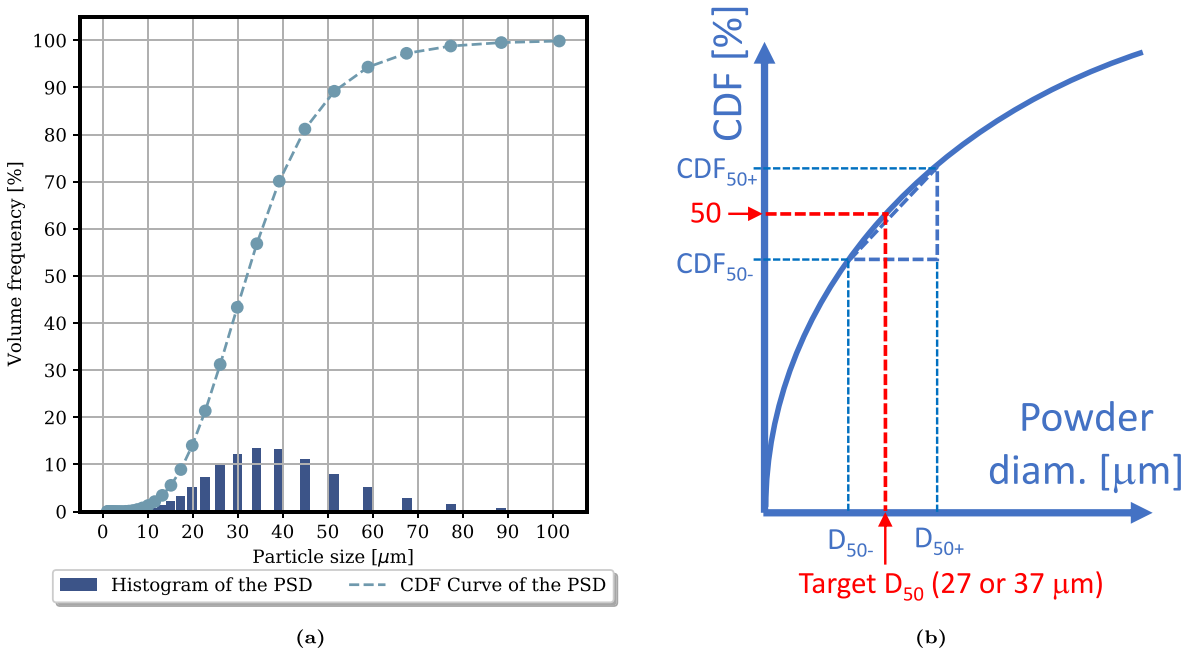


Fig. A.1. Explanation of the D50 estimation (a) PSD chart of the powder, (b) Calculating the resultant D50 for the constraint in Eq. (3).

by the higher noise (high porosity that is not included as a factor in the analysis) inherent with VED 2 that renders more factors insignificant (because of the lower F-values). Factor B is less significant than in the case with all VEDs considered. Meanwhile, the interaction BC is rendered insignificant for all QoIs.

Regression models for all the QoIs were constructed. Using the significant factors/interactions only as covariates showed a better performance than using the entire set of factors and interactions. This is shown as a reduction in the mean absolute percentage error (MAPE) applied to the leave-one-out cross-validation strategy.

CRediT authorship contribution statement

Peter Morcos: Writing – original draft, Methodology, Formal analysis. **David Shoukr:** Writing – original draft, Formal analysis, Conceptualization. **Tayler Sundermann:** Writing – original draft, Methodology. **Thomas Dobrowolski:** Methodology. **Nicholas Barta:** Methodology, Formal analysis. **Jayesh R. Jain:** Methodology, Formal analysis, Conceptualization. **Raymundo Arróyave:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Ibrahim Karaman:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Alaa Elwany:** Writing – review & editing, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge the support of the Baker Hughes Co. under Contract No. 2000480. Portions of this research were conducted in Baker Hughes facilities. The authors would like to acknowledge the invaluable insights and support from Chad Yates, Dr. Satya Ganti, and Dr. Wei Chen from Baker Hughes Company. The authors would like to acknowledge Dr. Brady Bulter and Dr. James Paramore from the Army Research Laboratory South and Rodney Inmon, Dr. Wilson Serem, Dr. Bryan Tomlin, and Dr Kadri C. Atli from the Texas A&M University for their help and insights during powder characterization and mechanical testing. Finally, the authors would like to acknowledge the hard work done by the undergraduate students in the powder preparation process; Andrew Yang, Zachary Butterfield, Dylan Wickersheim, Shiv Patel, Christian Valentine, and Ethan Weinberg without which this study would not have been possible.

Appendix A. Median particle size (D50) level control

An optimization model was developed and implemented for each powder batch to calculate the required mixing ratio of each sieved powder cut. For analysis, three PSD samples of 10 g each are collected from each powder cut. Each cut represents a portion of the entire weight of the sieved powder. Thus, the amount of powder that needs to be sieved to produce a certain amount of a certain cut 'i', $w(X_i)$, can be calculated as the reciprocal of the cut weight fraction, WF_i, multiplied by the required cut weight, X_i, as follows:

$$w(X_i) = X_i \left(\frac{1}{WF_i} \right) \tag{2}$$

These PSDs are then fed into an optimization algorithm to determine the volume fraction of each cut to be mixed such that the overall amount of powder needed is minimized while maintaining the design constraints. This optimization model has the following format:

Objective function

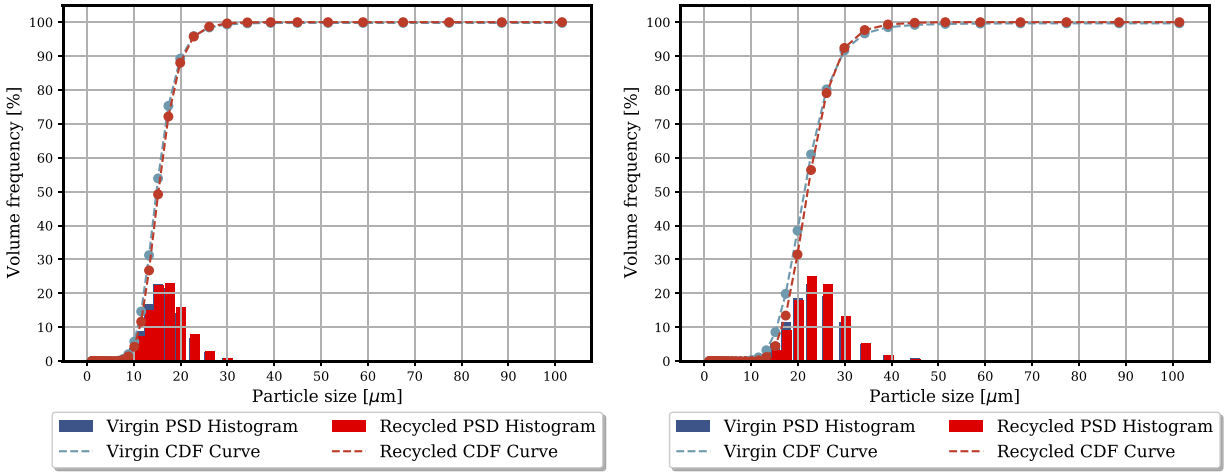
$$Minimize \quad w = max(w(X_1), w(X_2), \ldots, w(X_n)) \tag{3}$$

Subject to:

$$D_{50-} + \left(\frac{50 - CDF_{50-}}{CDF_{50+} - CDF_{50-}} \right) * (D_{50+} - D_{50-}) = D_{50,i} \tag{4}$$

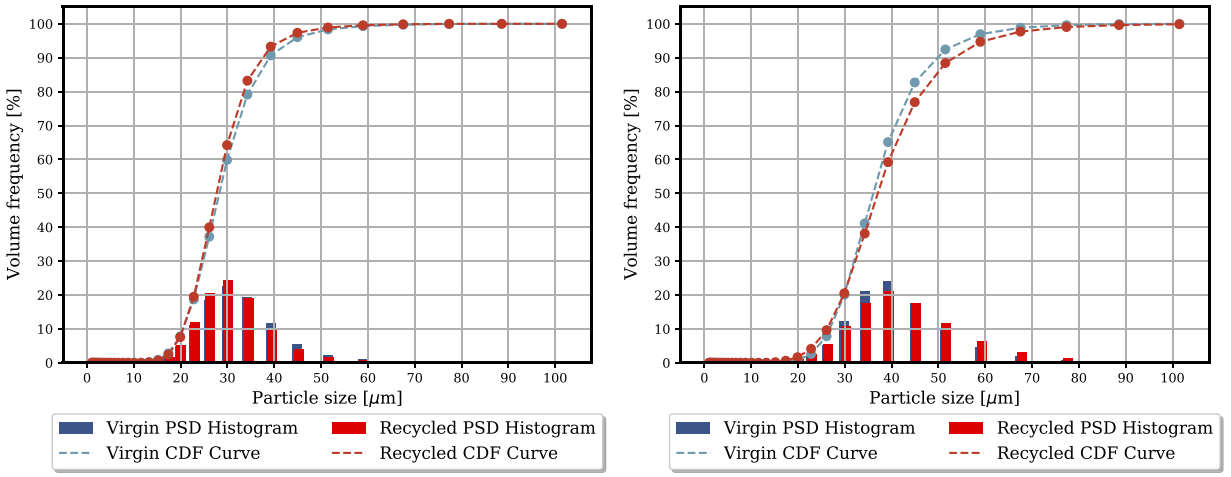
$$\sum_{i=1}^{n} X_i = 1 \tag{5}$$

$$0 < X_i < 1; \qquad X_i \neq X_{<20\ \mu m} \tag{6}$$



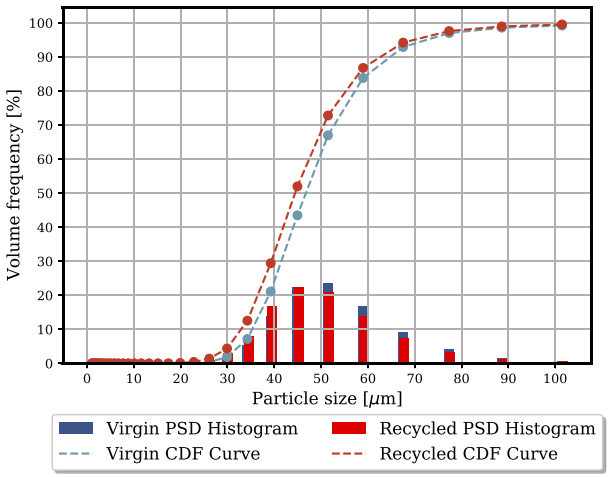
(a) Virgin D50 = 14.82, Recycled D50 = 15.25

(b) Virgin D50 = 21.35, Recycled D50 = 22.02



(c) Virgin D50 = 28.2, Recycled D50 = 27.62

(d) Virgin D50 = 36.03, Recycled D50 = 36.98



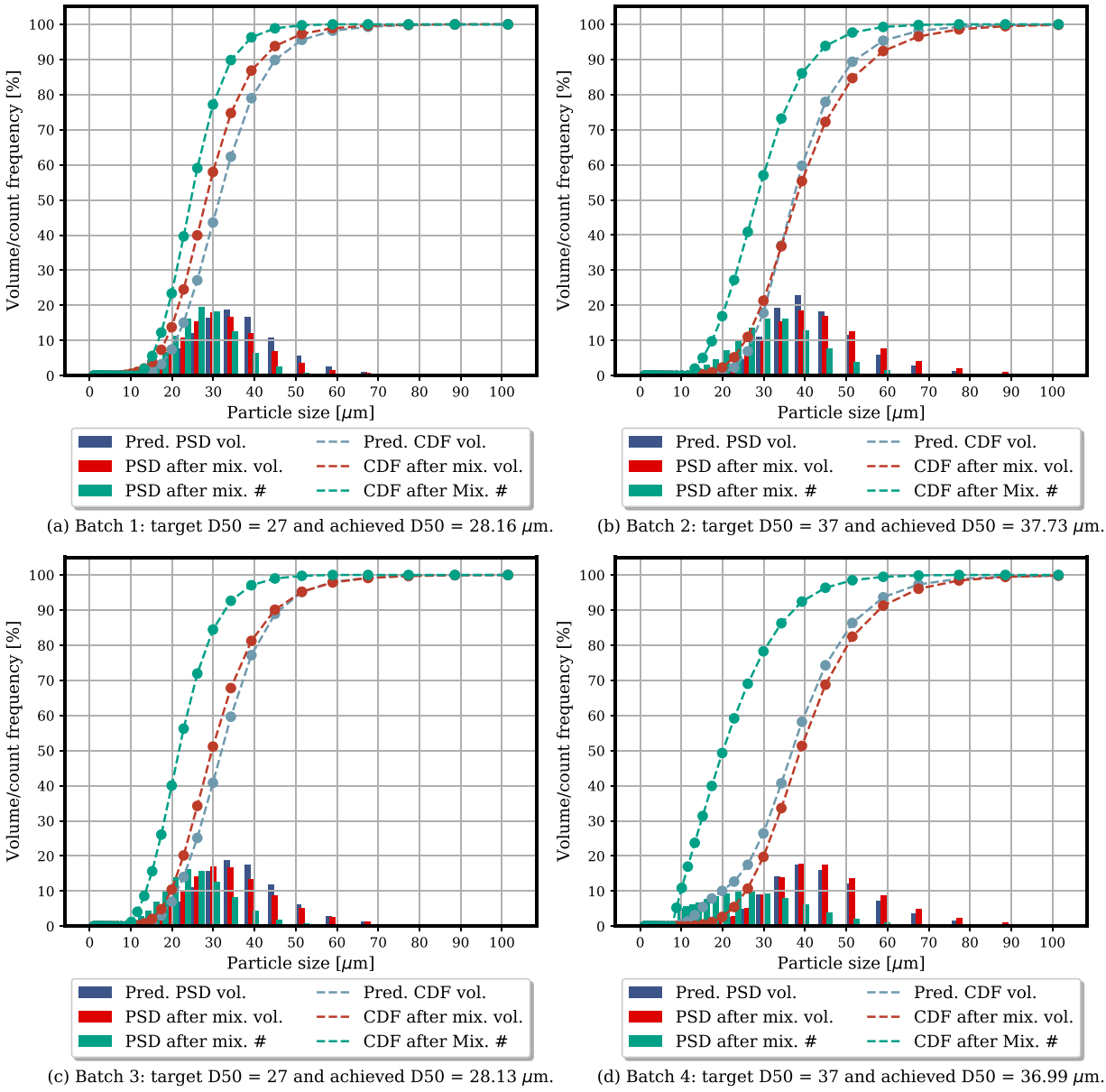
(e) Virgin D50 = 46.65, Recycled D50 = 44.4

Fig. A.2. Initial PSDs for the different sieving cuts (a) 0:20 μm, (b) 20:25 μm, (c) 25:32 μm, (d) 32:45 μm, and (e) 45:75 μm.

$$0.1 < X_{<20\ \mu m} < 1 \quad \text{for the full range particle size (Factor B2)} \tag{7}$$

Fig. A.1 shows an example of implementing the optimization model in calculating the D50 of the PSD for the initial virgin powder with D50 = 32.02 μm. As the PSD is measured at certain particle sizes, see Fig. A.1(a), the exact D50 value cannot be calculated. Instead, an approximate estimate is calculated using linear interpolation using Eq. (4). The two particle sizes where the D50 lies in between are D_{50-} and D_{50+} for which the corresponding cumulative amounts of powder are CDF_{50-} and CDF_{50+}, respectively. An illustration of the D50 calculation is shown in Fig. A.1(b).

The optimization problem is solved under the constraints of the D50 value and the lower and upper bounds of the weight fraction (decision variables) between (0, 1). Meanwhile, for the decision variable $X_{<20\ \mu m}$, $X_{<20\ \mu m} = 0$ for range 20:75 μm and has a lower bound of 0.1 for range



(a) Batch 1: target D50 = 27 and achieved D50 = 28.16 μm.

(b) Batch 2: target D50 = 37 and achieved D50 = 37.73 μm.

(c) Batch 3: target D50 = 27 and achieved D50 = 28.13 μm.

(d) Batch 4: target D50 = 37 and achieved D50 = 36.99 μm.

Fig. A.3. Predicted vs. actual PSD for virgin powder for (a) batch 1, (b) batch 2, (c) batch 3, and (d) batch 4.

0:75 μm. This forced higher lower bound is added to ensure that at least 10% of the powder, by volume, is of the finer particle sizes.

The initial PSDs for each powder cut are shown in Fig. A.2. It can be seen that the sieving process is working systematically for both powder types, virgin and recycled. Moreover, the PSD for each cut contains some powder coarser than the largest sieve size and finer than the smallest sieve size. This can be due to a lack of sphericity and the need for extra time for sieving, which is highly time-consuming and infeasible. To deal with that, the algorithm combines the different cuts based on the constituting particle sizes, i.e., when two cuts are combined, their histogram bin heights are multiplied by their corresponding weight fractions and then summed together. The predicted PSDs (by volume) for each batch as well as the actual PSDs (by volume as well as by count) after mixing for the different cuts are shown in Fig. A.3 for the virgin powder and Fig. A.4 for the recycled powder. Both figures show great conformity between the predicted and the mixing PSDs (by volume).

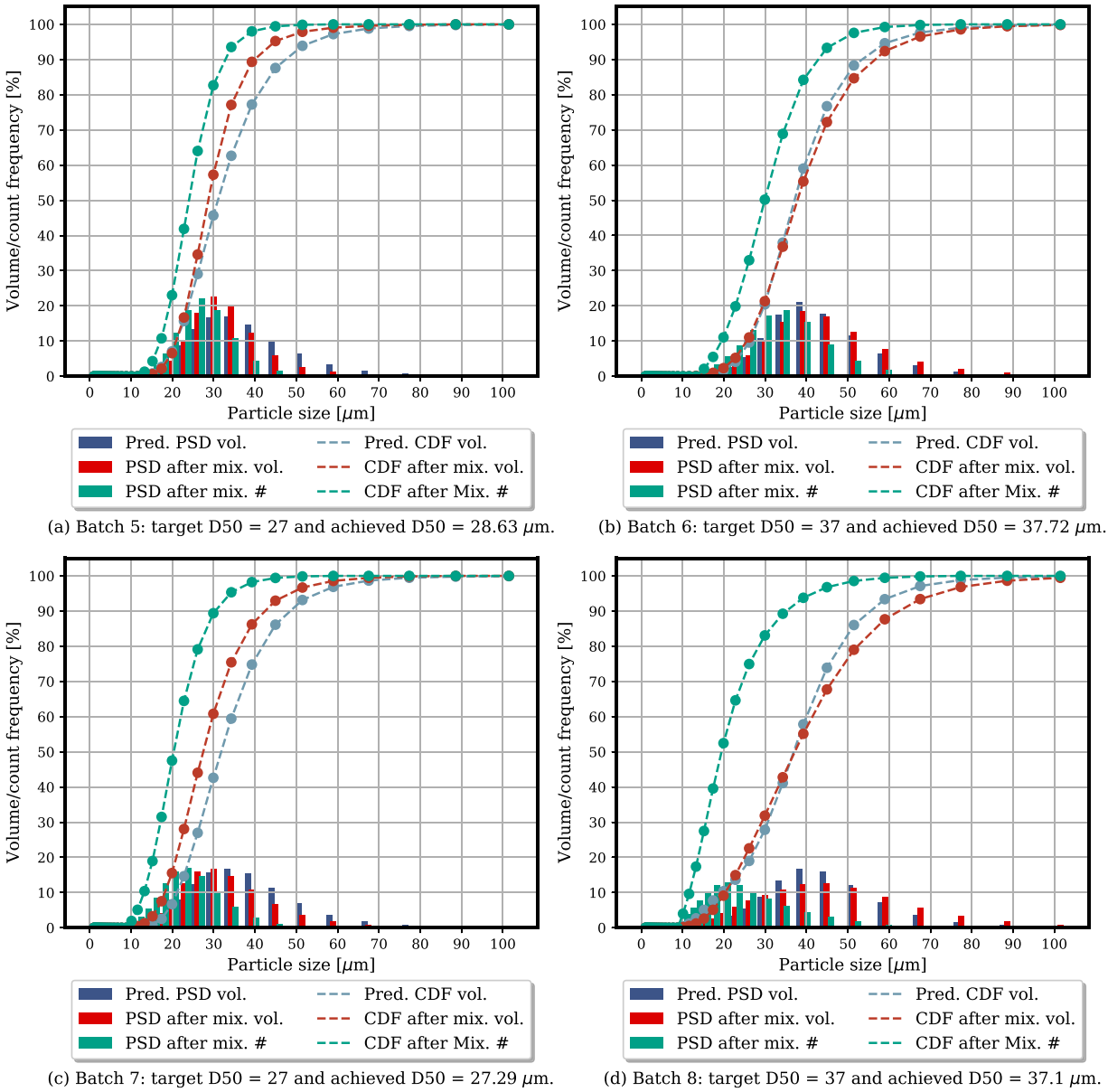
Appendix B. Analysis of variance (ANOVA)

The analysis of variance technique is used to determine the significance of the variability of the factor/interaction levels. It uses the sum of the squared differences between the factor level means and the grand mean of the QoI to determine the variability (variance) inherent because of these factor levels [44]. This variability (variance) is then compared to the variability of the error (noise) to determine its significance. The general idea is that if the factor/interaction variability is comparable to the error variability, then this variability could simply be due to the noise in the system. The main hypotheses to be tested by ANOVA are:

The null hypothesis, H_0 : $\mu_1 = \mu_2$

The alternative hypothesis, H_1 : $\mu_1 \neq \mu_2$

where μ_1 and μ_2 are the means of the two levels of each factor. The null hypothesis states that there is no difference in the factor-level means. On the other hand, the alternative hypothesis suggested that at least one of the factor-level means does not satisfy this condition. In other



(a) Batch 5: target D50 = 27 and achieved D50 = 28.63 μm.

(b) Batch 6: target D50 = 37 and achieved D50 = 37.72 μm.

(c) Batch 7: target D50 = 27 and achieved D50 = 27.29 μm.

(d) Batch 8: target D50 = 37 and achieved D50 = 37.1 μm.

Fig. A.4. Predicted vs. actual PSD for recycled powder for (a) batch 5, (b) batch 6, (c) batch 7, and (d) batch 8.

words, if the means of the levels of a factor are all equal (H_0), then the variation of this factor does not affect the QoI. Otherwise, this factor has a significant effect. The differences in the means are calculated using the sum of squares (SS) formulae. For the total sum of squares (SS_T) is

$$SS_T = SS_{model} + SS_e = \sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{..})^2 \tag{8}$$

and for the model (SS_{model}) is

$$SS_{model} = n\sum_{k=1}^{b}\sum_{l=1}^{a_k}(\bar{y}_{l.} - \bar{y}_{..})^2 \tag{9}$$

while for each factor k (SS_k) is

$$SS_{model} = \sum_{l=1}^{a_k}(\bar{y}_{l.} - \bar{y}_{..})^2 \tag{10}$$

where, SS_{model} is the sum of all the sums of squares of the main/interaction effects, y_{ij} is the response of the jth replicate of treatment (factor-level combination) i, $\bar{y}_{..}$ is the grand mean of all responses, $\bar{y}_{l.}$ is the mean response of level l of factor k, a is the number of treatments, n is the number of replicates, b is the number of the factors, and a_k is the number of levels of factor k.

Hence, $SS_{model} = \sum_{k=1}^{b} SS_k$. SS_k is the sum of squares of factor k. The sum of the squares of each factor is then divided by the degrees of freedom to calculate an unbiased measure of the variability (variance). Then, the mean sum of the squares of the factor is divided by (compared to) the mean sum of the squares of the error to get the F-value. Finally, the F-value and the degrees of freedom are used to estimate the p-value from the corresponding F-distribution. The factors that have a p-value $< \alpha$ (level of significance) are statistically significant (H_0 is rejected).

Appendix C. Regression models

Statistical regression is a method to predict an unknown QoI (dependent variable) from the values of the affecting factors (independent variables) based on the existing data. A typical linear regression model consists of an intercept (the value of the QoI if all the factors are equal to zero) and the factors/interactions multiplied by their coefficients

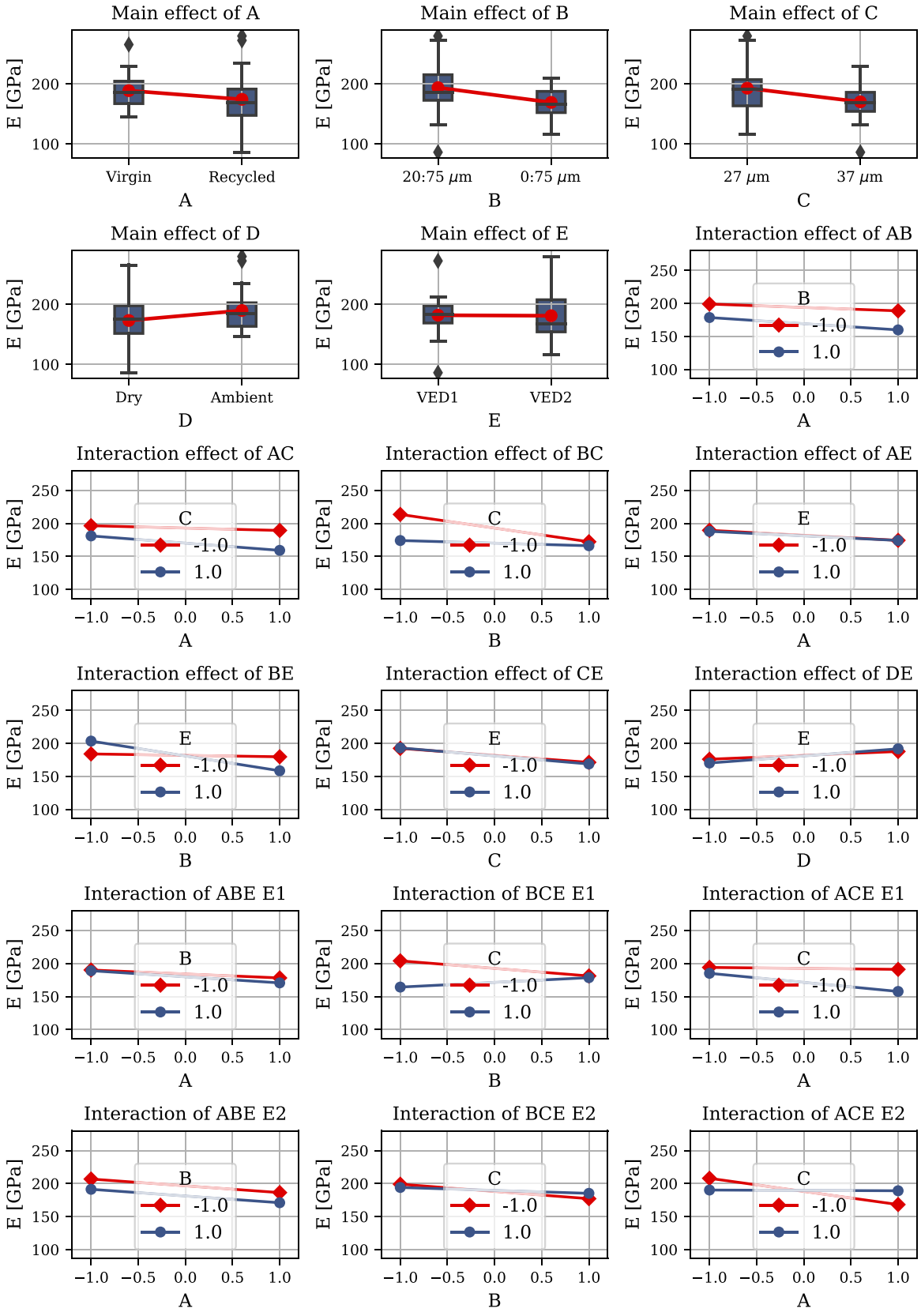


Fig. D.1. Effect plots of the modulus of elasticity QoI.

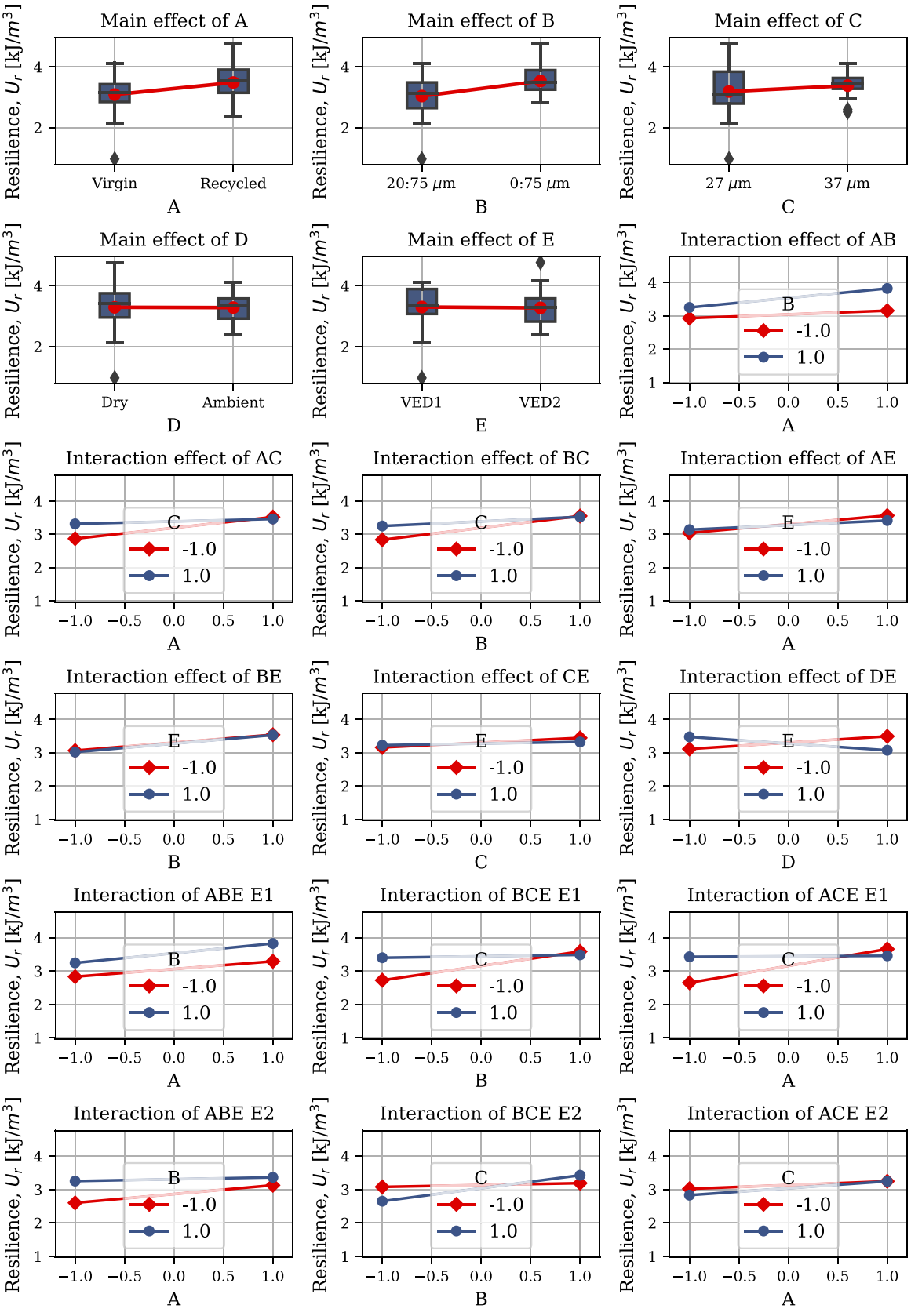


Fig. D.2. Effect plots of the resilience QoI.

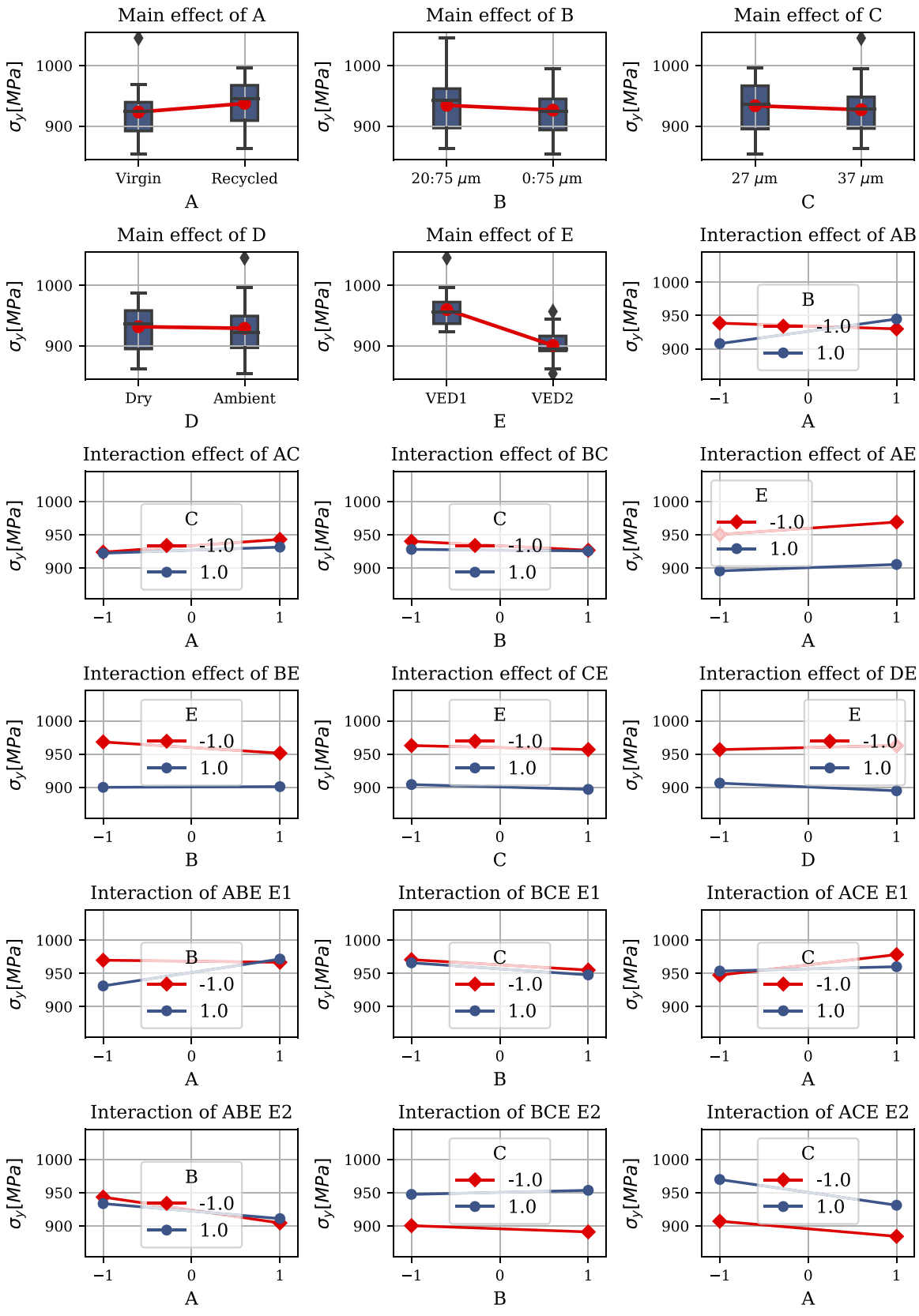


Fig. D.3. Effect plots of the yield strength QoI.

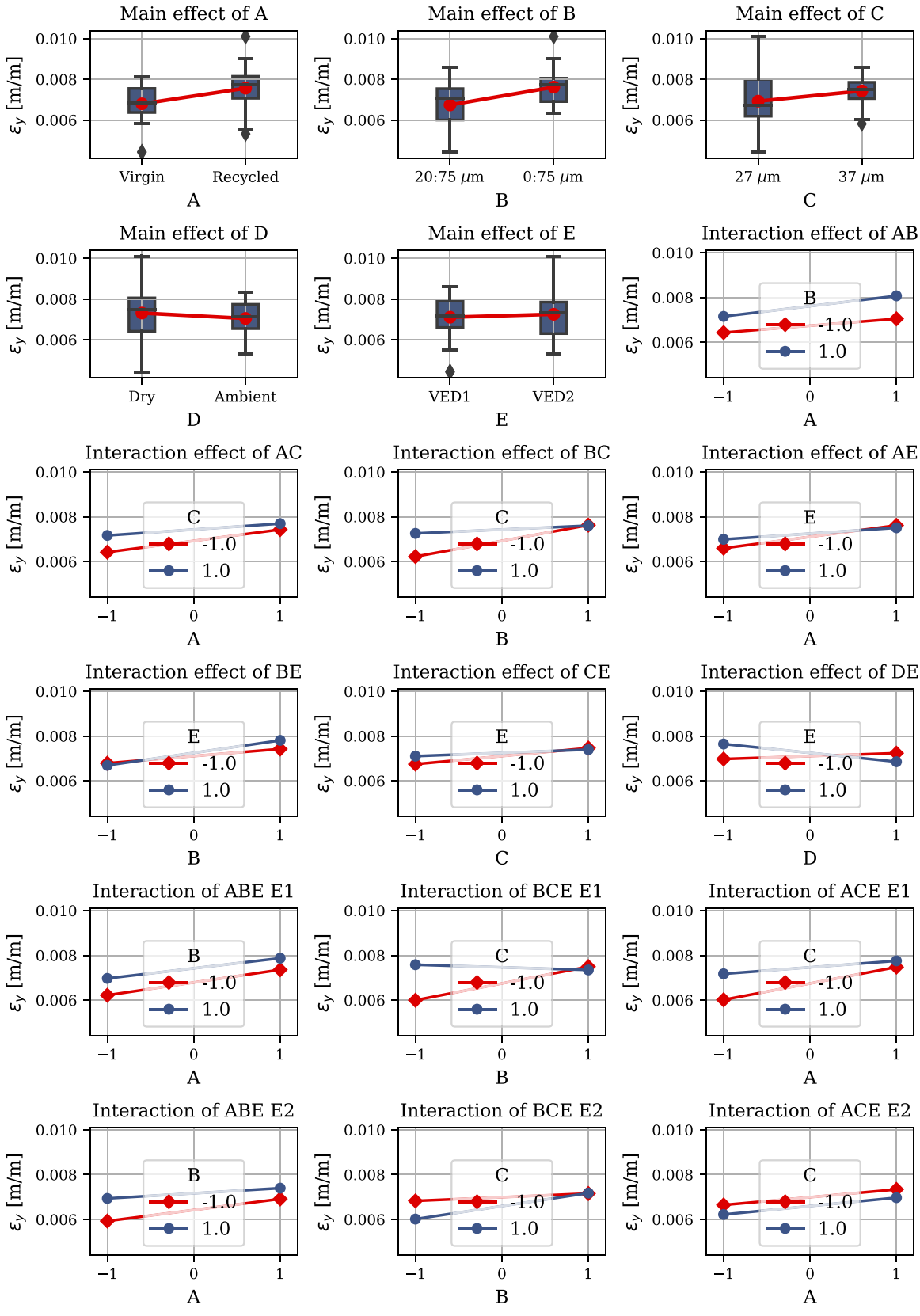


Fig. D.4. Effect plots of the yield strain QoI.

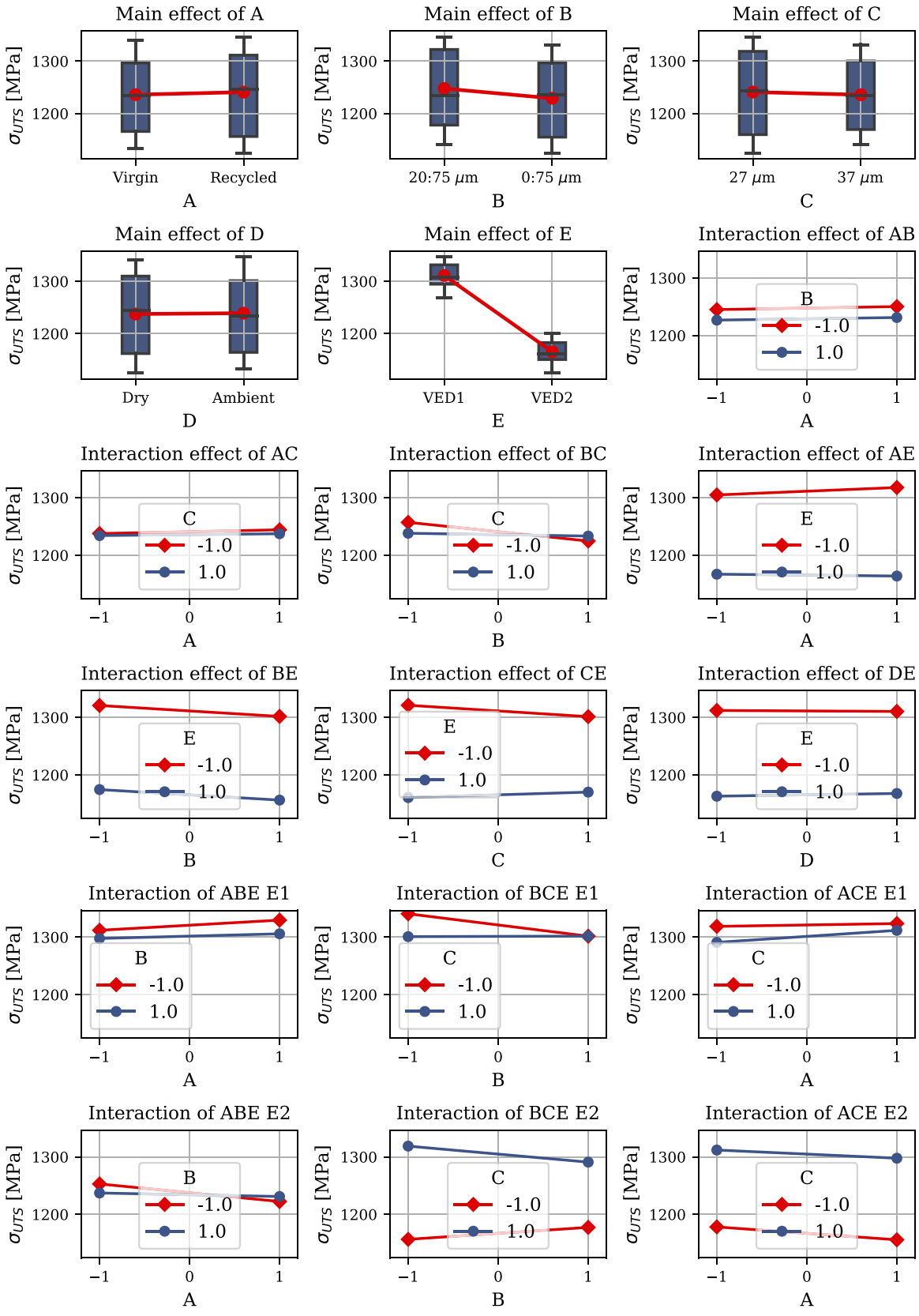


Fig. D.5. Effect plots of the ultimate tensile strength QoI.

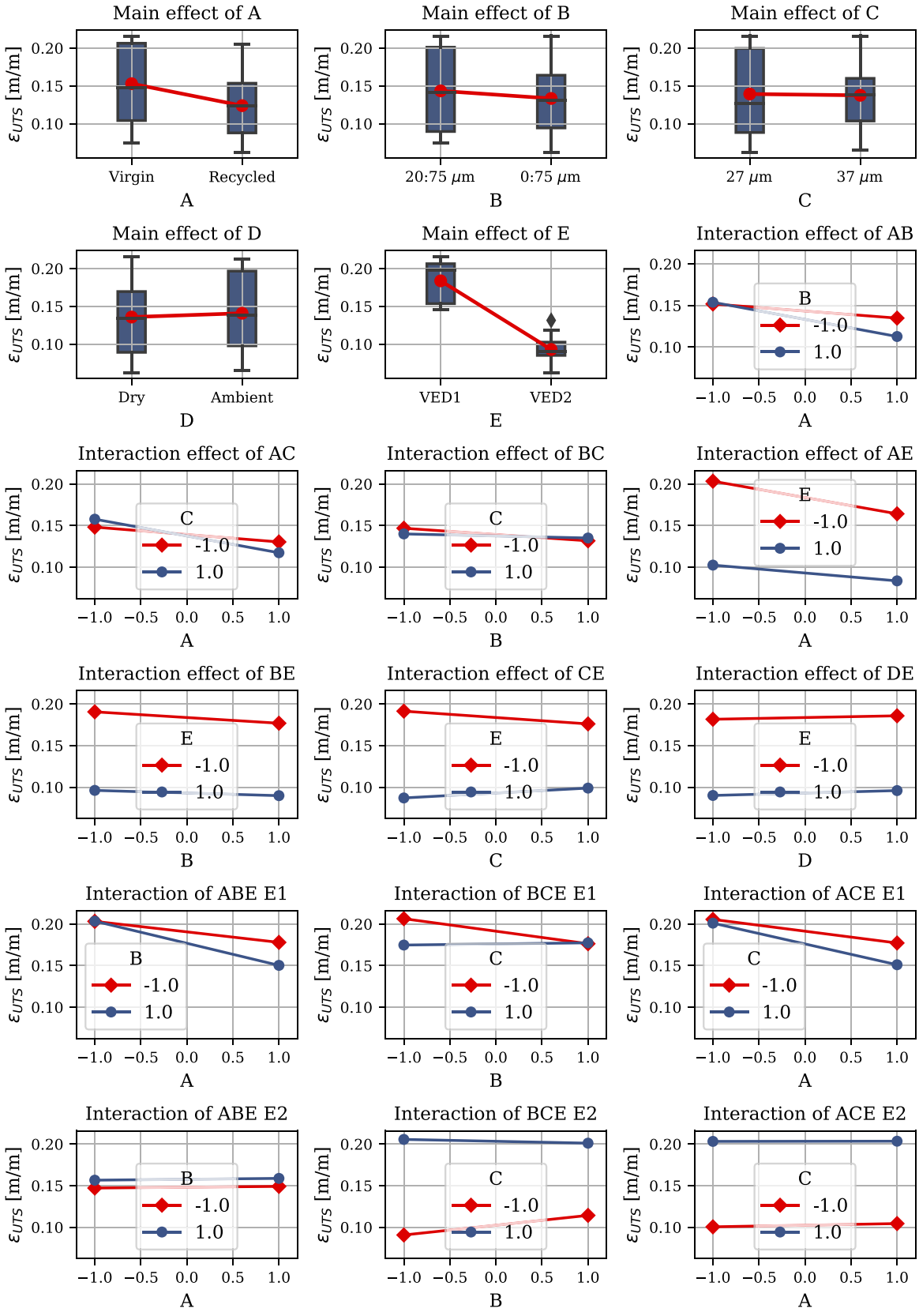


Fig. D.6. Effect plots of the strain at the occurrence of the ultimate tensile strength QoI.

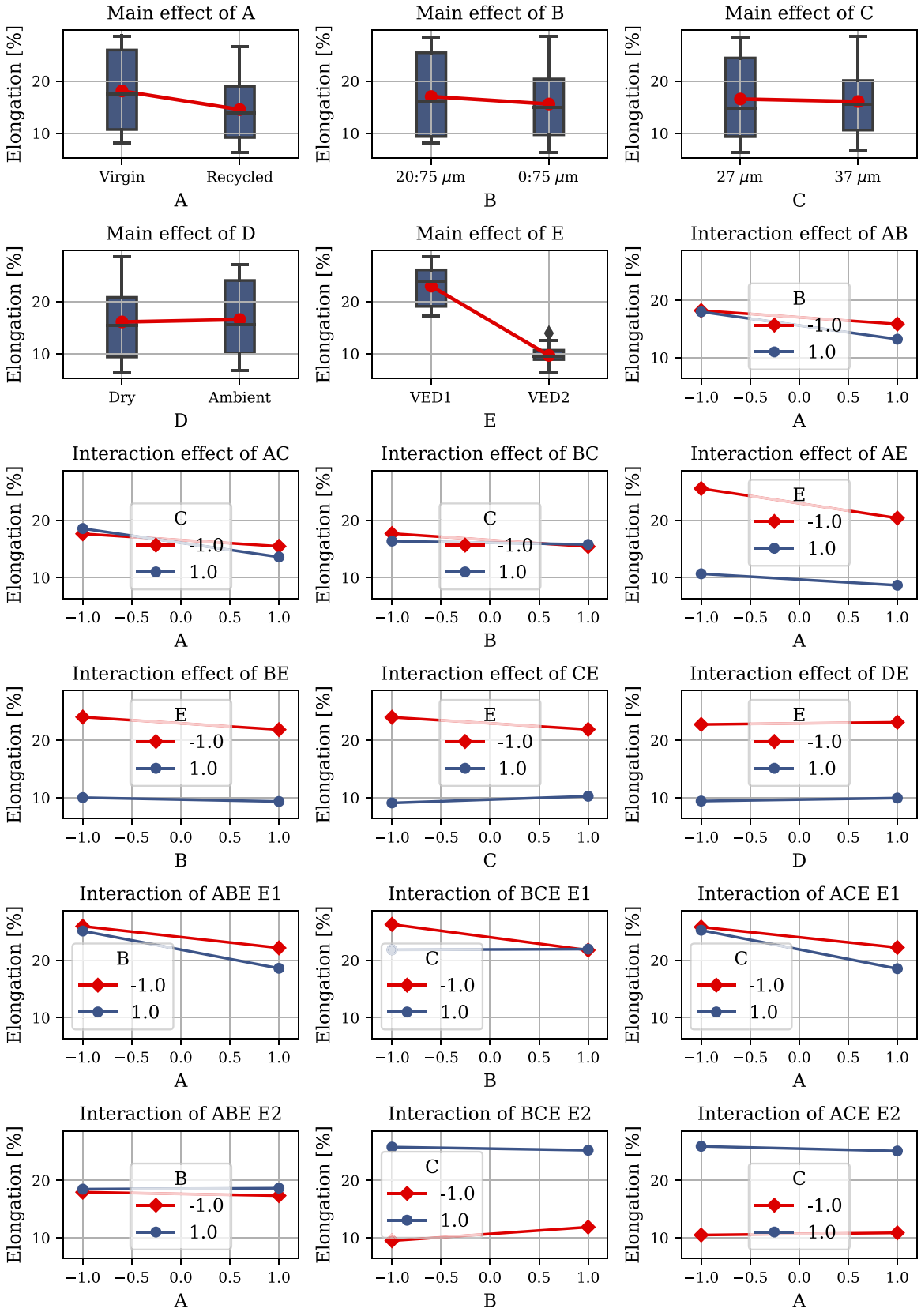


Fig. D.7. Effect plots of the elongation at fracture QoI.

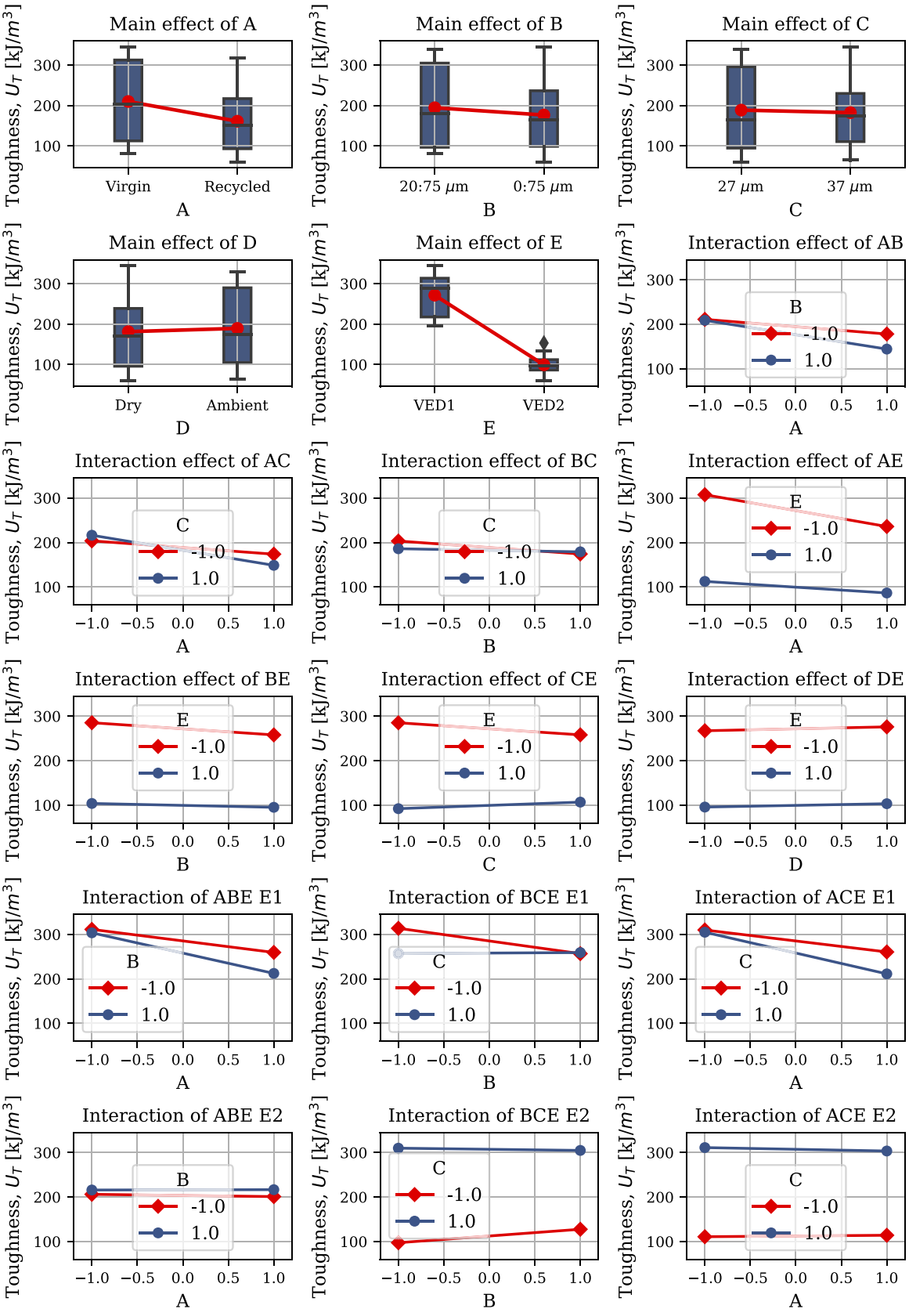


Fig. D.8. Effect plots of the toughness QoI.

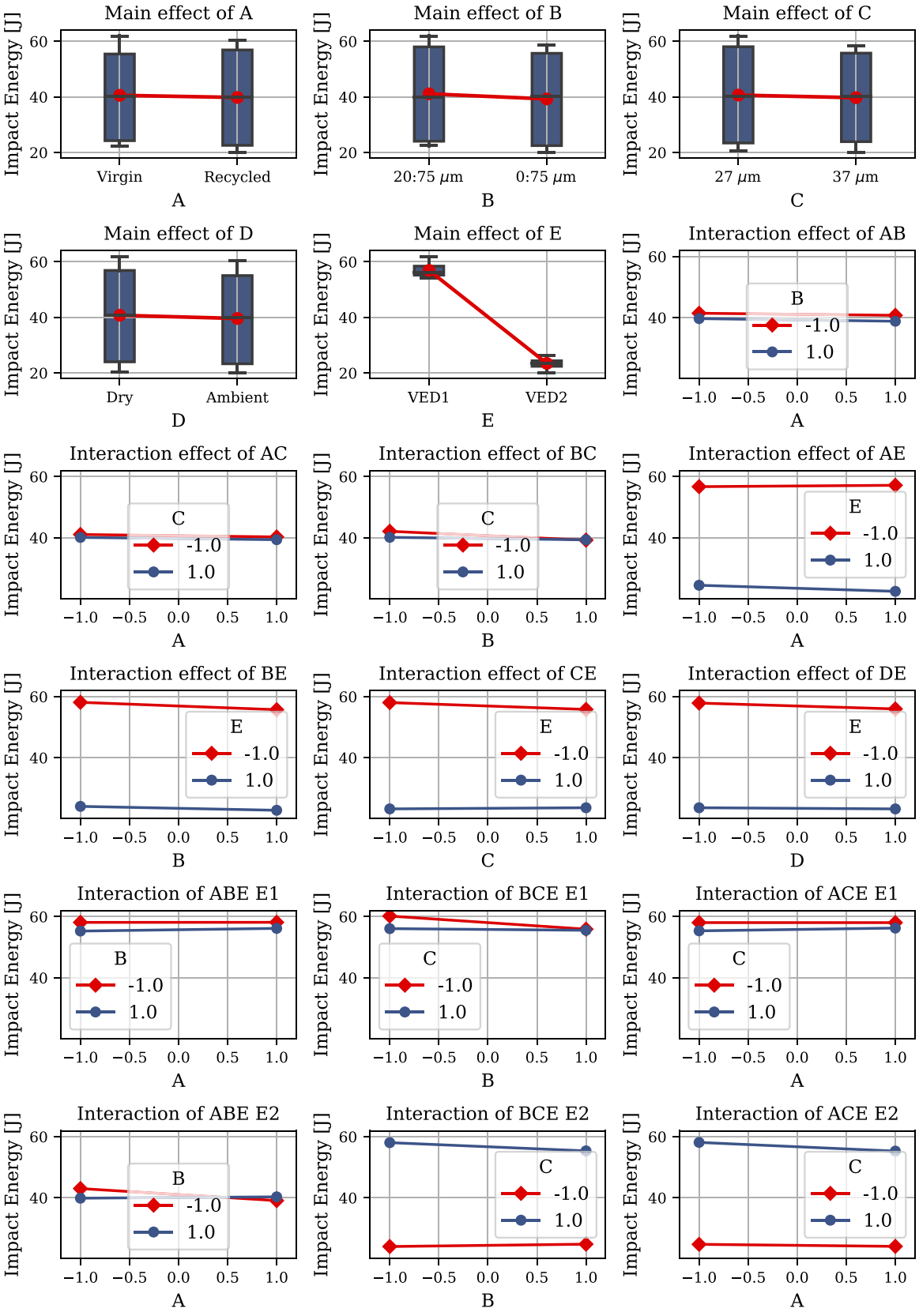


Fig. D.9. Effect plots of the Charpy impact energy QoI.

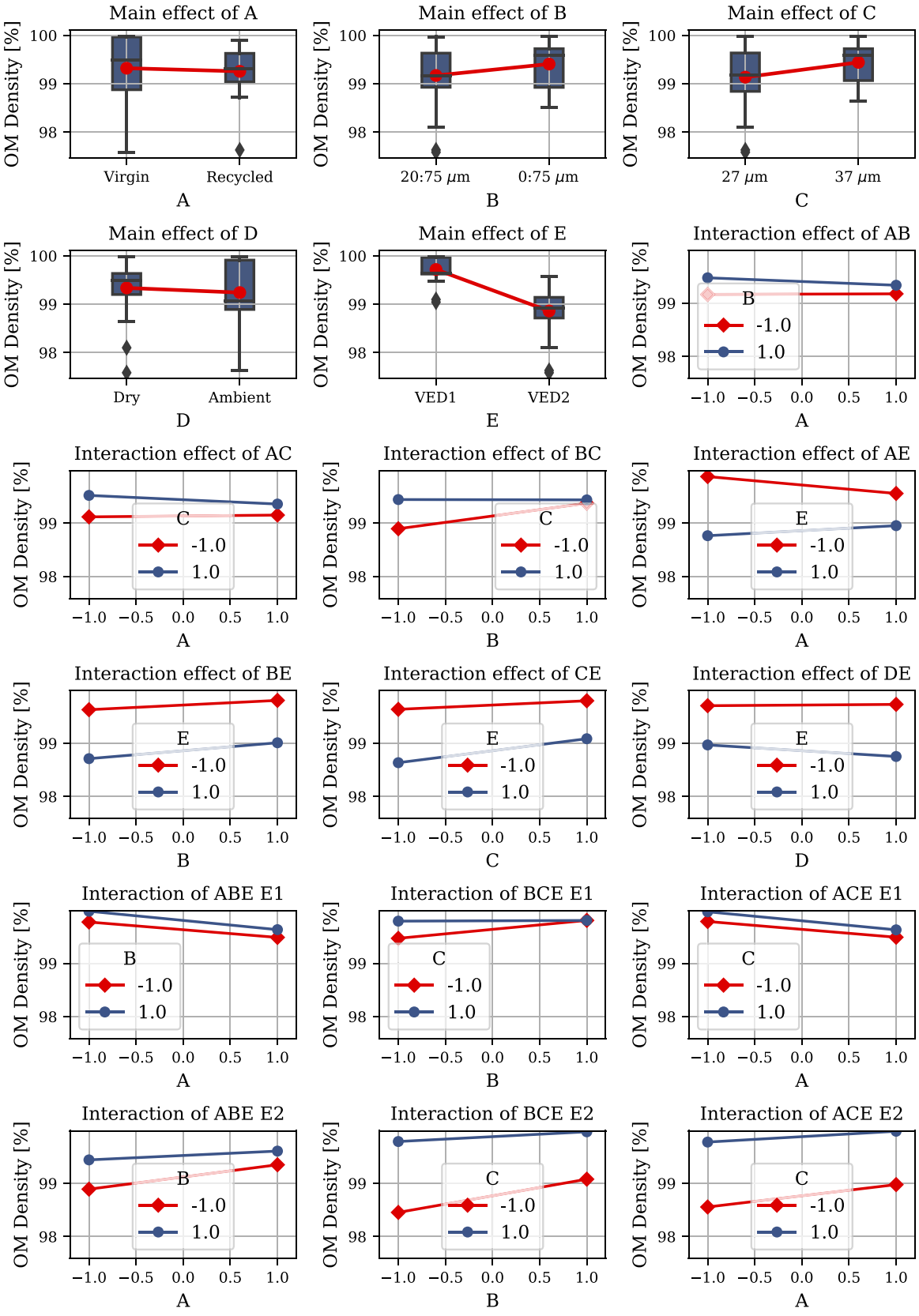


Fig. D.10. Effect plots of the OM densification QoI.

according to the following equation:

$$y = F(x) = \beta_0 + \sum_{i \in Q} \beta_i * x_i; \quad Q = \{A, B, C, \ldots, BCE\} \tag{11}$$

The intercept and coefficients (β_0 and β_i) are calculated according to the ordinary least squares (OLS) method [52]. The OLS method estimates the coefficient values such that the squared differences between the observed and the predicted QoI values are minimized

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\| \tag{12}$$

where, $\hat{\boldsymbol{\beta}}$ is the vector of the estimated coefficients, $\|\cdot\|$ is the standard L^2 norm and $\boldsymbol{X}$ is the design matrix. To avoid over-fitting, only the significant terms for each QoI in its corresponding ANOVA table are included in the design matrix $\boldsymbol{X}$. The term $\boldsymbol{X}\boldsymbol{\beta}$ represents the predicted values while $\boldsymbol{y}$ is the corresponding observed values vector. It can be easily proven that the minimum OLS estimator for $\hat{\boldsymbol{\beta}}$ is [51]

$$\hat{\boldsymbol{\beta}} = \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} \tag{13}$$

To calculate the effect of the categorical variable levels of factors A, B, and D, one can simply plug −1 or +1 into the equation corresponding to the lower and upper levels of each factor. Meanwhile, the implementation of the continuous variables C and E is straightforward.

Appendix D. D-effect plots

The main and interaction effect plots of each QoI are presented in Figs. D.1–D.10. Only the interactions considered in the ANOVA analysis are included in these figures. The three-factor interactions are three-dimensional, so they are presented here over two subplots.

References

[1] O. Adegoke, J. Andersson, H. Brodin, R. Pederson, Review of laser powder bed fusion of Gamma-prime-strengthened nickel-based superalloys, Metals 10 (8) (2020) http://dx.doi.org/10.3390/met10080996.

[2] M. Bonesso, P. Rebesan, C. Gennari, S. Mancin, R. Dima, A. Pepato, I. Calliari, Effect of particle size distribution on laser powder bed fusion manufacturability of copper, BHM Berg- und Hüttenmännische Monatshefte 166 (5) (2021) 256–262, http://dx.doi.org/10.1007/s00501-021-01107-0.

[3] T. Ronneberg, C.M. Davies, P.A. Hooper, Revealing relationships between porosity, microstructure and mechanical properties of laser powder bed fusion 316l stainless steel through heat treatment, Mater. Des. 189 (2020) 108481, http://dx.doi.org/10.1016/j.matdes.2020.108481.

[4] J. Zhang, B. Hu, Y. Zhang, X. Guo, L. Wu, H.-Y. Park, J.-H. Lee, Y.-G. Jung, Comparison of virgin and reused 15-5 PH stainless steel powders for laser powder bed fusion process, Prog. Addit. Manuf. 3 (1–2) (2018) 11–14, http://dx.doi.org/10.1007/s40964-018-0038-2.

[5] H. Irrinki, J.S.D. Jangam, S. Pasebani, S. Badwe, J. Stitzel, K. Kate, O. Gulsoy, S.V. Atre, Effects of particle characteristics on the microstructure and mechanical properties of 17-4 PH stainless steel fabricated by laser-powder bed fusion, Powder Technol. 331 (2018) 192–203, http://dx.doi.org/10.1016/j.powtec.2018.03.025.

[6] P.A. Kuznetsov, I.V. Shakirov, A.S. Zukov, V.V. Bobyr', M.V. Starytsin, Effect of particle size distribution on the structure and mechanical properties in the process of laser powder bed fusion, J. Phys. Conf. Ser. 1758 (1) (2021) 012021, http://dx.doi.org/10.1088/1742-6596/1758/1/012021.

[7] S.D. Nath, G. Gupta, M. Kearns, O. Gulsoy, S.V. Atre, Effects of layer thickness in laser-powder bed fusion of 420 stainless steel, Rapid Prototyp. J. 26 (7) (2020) 1197–1208, http://dx.doi.org/10.1108/rpj-10-2019-0279.

[8] P. Wang, A. Salandari-Rabori, Q. Dong, V. Fallah, Effect of input powder attributes on optimized processing and as-built tensile properties in laser powder bed fusion of AlSi10Mg alloy, J. Manuf. Process. 64 (2021) 633–647, http://dx.doi.org/10.1016/j.jmapro.2021.02.024.

[9] G. Shanbhag, E. Wheat, S. Moylan, M. Vlasea, Effect of specimen geometry and orientation on tensile properties of Ti-6Al-4V manufactured by electron beam powder bed fusion, Addit. Manuf. 48 (2021) 102366, http://dx.doi.org/10.1016/j.addma.2021.102366.

[10] L. Xue, K. Atli, C. Zhang, N. Hite, A. Srivastava, A. Leff, A. Wilson, D. Sharar, A. Elwany, R. Arroyave, I. Karaman, Laser powder bed fusion of defect-free NiTi shape memory alloy parts with superior tensile superelasticity, Acta Mater. 229 (2022) 117781, http://dx.doi.org/10.1016/j.actamat.2022.117781.

[11] L. Cordova, T. Bor, M. de Smit, S. Carmignato, M. Campos, T. Tinga, Effects of powder reuse on the microstructure and mechanical behaviour of Al–Mg–Sc–Zr alloy processed by laser powder bed fusion (LPBF), Addit. Manuf. 36 (2020) 101625, http://dx.doi.org/10.1016/j.addma.2020.101625.

[12] N.A. Zumdick, L. Jauer, L.C. Kersting, T.N. Kutz, J.H. Schleifenbaum, D. Zander, Additive manufactured WE43 magnesium: A comparative study of the microstructure and mechanical properties with those of powder extruded and as-cast WE43, Mater. Charact. 147 (2019) 384–397, http://dx.doi.org/10.1016/j.matchar.2018.11.011.

[13] P. Morcos, A. Elwany, I. Karaman, R. Arróyave, Additive manufacturing of pure tungsten and tungsten-based alloys, J. Mater. Sci. 57 (21) (2022) 9769–9806.

[14] T. De Terris, O. Castelnau, Z. Hadjem-Hamouche, H. Haddadi, V. Michel, P. Peyre, Analysis of as-built microstructures and recrystallization phenomena on inconel 625 alloy obtained via laser powder bed fusion (L-PBF), Metals 11 (4) (2021) http://dx.doi.org/10.3390/met11040619.

[15] B.B. Ravichander, A. Amerinatanzi, N. Shayesteh Moghaddam, Study on the effect of powder-bed fusion process parameters on the quality of as-built IN718 parts using response surface methodology, Metals 10 (9) (2020) http://dx.doi.org/10.3390/met10091180.

[16] B. Sutton, E. Herderick, R. Thodla, M. Ahlfors, A. Ramirez, Heat treatment of alloy 718 made by additive manufacturing for oil and gas applications, JOM 71 (3) (2019) 1134–1143.

[17] S. Sanchez, P. Smith, Z. Xu, G. Gaspard, C.J. Hyde, W.W. Wits, I.A. Ashcroft, H. Chen, A.T. Clare, Powder bed fusion of nickel-based superalloys: A review, Int. J. Mach. Tools Manuf. 165 (2021) 103729, http://dx.doi.org/10.1016/j.ijmachtools.2021.103729.

[18] J.P. Oliveira, A. LaLonde, J. Ma, Processing parameters in laser powder bed fusion metal additive manufacturing, Mater. Des. 193 (2020) 108762.

[19] Q. Chen, Y. Zhao, S. Strayer, Y. Zhao, K. Aoyagi, Y. Koizumi, A. Chiba, W. Xiong, A.C. To, Elucidating the effect of preheating temperature on melt pool morphology variation in Inconel 718 laser powder bed fusion via simulation and experiment, Addit. Manuf. 37 (2021) 101642, http://dx.doi.org/10.1016/j.addma.2020.101642.

[20] D. Shoukr, P. Morcos, T. Sundermann, T. Dobrowolski, C. Yates, J.R. Jain, R. Arróyave, I. Karaman, A. Elwany, Influence of layer thickness on the printability of nickel alloy 718: A systematic process optimization framework, Addit. Manuf. (2023) 103646.

[21] A. Raza, C. Pauzon, E. Hryha, A. Markström, P. Forêt, Spatter oxidation during laser powder bed fusion of Alloy 718: Dependence on oxygen content in the process atmosphere, Addit. Manuf. 48 (2021) 102369, http://dx.doi.org/10.1016/j.addma.2021.102369.

[22] C. Pauzon, A. Markström, S.D.-L. Goff, E. Hryha, Effect of the process atmosphere composition on alloy 718 produced by laser powder bed fusion, Metals 11 (8) (2021) 1254, http://dx.doi.org/10.3390/met11081254.

[23] S.A. Farzadfar, M.J. Murtagh, N. Venugopal, Impact of IN718 bimodal powder size distribution on the performance and productivity of laser powder bed fusion additive manufacturing process, Powder Technol. 375 (2020) 60–80, http://dx.doi.org/10.1016/j.powtec.2020.07.092.

[24] S.E. Brika, M. Letenneur, C.A. Dion, V. Brailovski, Influence of particle morphology and size distribution on the powder flowability and laser powder bed fusion manufacturability of Ti-6Al-4V alloy, Addit. Manuf. 31 (2020) 100929, http://dx.doi.org/10.1016/j.addma.2019.100929.

[25] K. Gruber, I. Smolina, M. Kasprowicz, T. Kurzynowski, Evaluation of inconel 718 metallic powder to optimize the reuse of powder and to improve the performance and sustainability of the laser powder bed fusion (LPBF) process, Materials 14 (6) (2021) 1538, http://dx.doi.org/10.3390/ma14061538.

[26] L. Cordova, M. Campos, T. Tinga, Revealing the effects of powder reuse for selective laser melting by powder characterization, JOM 71 (3) (2019) 1062–1072, http://dx.doi.org/10.1007/s11837-018-3305-2.

[27] C. Rock, C. Ledford, M. Garcia-Avila, H. West, V.M. Miller, M. Pankow, R. Dehoff, T. Horn, The influence of powder reuse on the properties of nickel super alloy ATI 718™ in laser powder bed fusion additive manufacturing, Metall. Mater. Trans. B 52 (2) (2021) 676–688, http://dx.doi.org/10.1007/s11663-020-02040-2.

[28] B.A. Hann, Powder reuse and its effects on laser based powder fusion additive manufactured alloy 718, SAE Int. J. Aerosp. 9 (2) (2016) 209–214.

[29] B. Sartin, T. Pond, B. Griffith, W. Everhart, L. Elder, E. Wenski, C. Cook, D. Wieliczka, W. King, A. Rubenchik, et al., 316L powder reuse for metal additive manufacturing, in: 2017 International Solid Freeform Fabrication Symposium, University of Texas at Austin, 2017.

[30] L. Cordova, M. Campos, T. Tinga, Revealing the effects of powder reuse for selective laser melting by powder characterization, Jom 71 (3) (2019) 1062–1072.

[31] L. Cordova, T. Bor, M. de Smit, S. Carmignato, M. Campos, T. Tinga, Effects of powder reuse on the microstructure and mechanical behaviour of Al–Mg–Sc–Zr alloy processed by laser powder bed fusion (LPBF), Addit. Manuf. 36 (2020) 101625, http://dx.doi.org/10.1016/j.addma.2020.101625.

[32] API Standard 6ACRA, Age-hardened Nickel-based alloys for oil and gas drilling and production equipment, 2015.

[33] Q.B. Nguyen, M.L.S. Nai, Z. Zhu, C.-N. Sun, J. Wei, W. Zhou, Characteristics of inconel powders for powder-bed additive manufacturing, Engineering 3 (5) (2017) 695–700, http://dx.doi.org/10.1016/J.ENG.2017.05.012.

[34] Y. Bai, G. Wagner, C.B. Williams, Effect of particle size distribution on powder packing and sintering in binder jetting additive manufacturing of metals, J. Manuf. Sci. Eng. 139 (8) (2017) http://dx.doi.org/10.1115/1.4036640.

[35] D. Gu, M. Xia, D. Dai, On the role of powder flow behavior in fluid thermodynamics and laser processability of Ni-based composites by selective laser melting, Int. J. Mach. Tools Manuf. (2019).

[36] L. Cordova, T. Bor, M. de Smit, M. Campos, T. Tinga, Measuring the spreadability of pre-treated and moisturized powders for laser powder bed fusion, Addit. Manuf. 32 (2020) 101082, http://dx.doi.org/10.1016/j.addma.2020.101082.

[37] D. Powell, A.E. Rennie, L. Geekie, N. Burns, Understanding powder degradation in metal additive manufacturing to allow the upcycling of recycled powders, J. Clean. Prod. 268 (2020) 122077, http://dx.doi.org/10.1016/j.jclepro.2020.122077.

[38] ASTM International, E11-95: Standard specification for wire cloth and sieves for testing purposes, 2017.

[39] E. Baril, L.P. Lefebvre, Y. Thomas, Interstitial elements in titanium powder metallurgy: Sources and control, Powder Metall. 54 (3) (2011) 183–186, http://dx.doi.org/10.1179/174329011x13045076771759.

[40] Y. Sun, M. Aindow, R.J. Hebert, Comparison of virgin ti-6al-4v powders for additive manufacturing, Addit. Manuf. 21 (2018) 544–555, http://dx.doi.org/10.1016/j.addma.2018.02.011.

[41] H. Conrad, Effect of interstitial solutes on the strength and ductility of titanium, Prog. Mater. Sci. 26 (2–4) (1981) 123–403, http://dx.doi.org/10.1016/0079-6425(81)90001-3.

[42] C.J. Wu, M.S. Hamada, Experiments: Planning, Analysis, and Optimization, Vol. 552, John Wiley & Sons, 2011.

[43] G.E. Box, J.S. Hunter, W.G. Hunter, et al., Statistics for Experimenters: Design, Innovation, and Discovery, Vol. 2, Wiley-Interscience New York, 2005.

[44] D. Montgomery, Design and Analysis of Experiments, John Wiley & Sons, 2008.

[45] A. International, Standard test method for particle size distribution of metal powders and related compounds by light scattering, Standard ASTM B822-17, American Society for Testing and Materials, 100 Barr Harbor Drive, PO Box C700, West Conshohocken,PA 19428, USA, 2017.

[46] ASTM International, Standard test methods for tension testing of metallic materials, Standard ASTM E8/E8M-21, American Society for Testing and Materials, 100 Barr Harbor Drive, PO Box C700, West Conshohocken,PA 19428, USA, 2021.

[47] Standard, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428, USA.

[48] U. Ali, Y. Mahmoodkhani, S. Imani Shahabad, R. Esmaeilizadeh, F. Liravi, E. Sheydaeian, K.Y. Huang, E. Marzbanrad, M. Vlasea, E. Toyserkani, et al., On the measurement of relative powder-bed compaction density in powder-bed additive manufacturing processes, Mater. Des. 155 (2018) 495–501, http://dx.doi.org/10.1016/j.matdes.2018.06.030.

[49] L. Dowling, J. Kennedy, S. O'Shaughnessy, D. Trimble, A review of critical repeatability and reproducibility issues in powder bed fusion, Mater. Des. 186 (2020) 108346, http://dx.doi.org/10.1016/j.matdes.2019.108346.

[50] G. Jacob, A. Donmez, J. Slotwinski, S. Moylan, Measurement of powder bed density in powder bed fusion additive manufacturing processes, Meas. Sci. Technol. 27 (11) (2016) 115601, http://dx.doi.org/10.1088/0957-0233/27/11/115601.

[51] D.C. Montgomery, E.A. Peck, G.G. Vining, Introduction to Linear Regression Analysis, John Wiley & Sons, 2021.

[52] S. Seabold, J. Perktold, Statsmodels: Econometric and statistical modeling with python, in: Proceedings of the 9th Python in Science Conference, Vol. 57, Austin, TX, 2010, p. 61.